ELEMENTARY PROBABILITY

SECOND EDITION

David Stirzaker



CAMBRIDGE

more information - www.cambridge.org/9780521833448

4
Random Variables: Distribution and Expectation

I am giddy, expectation whirls me round.
William Shakespeare, Troilus and Cressida

4.1 Random Variables

In many experiments, outcomes are defined in terms of numbers (e.g., the number of heads in n tosses of a coin) or may be associated with numbers, if we so choose. In either case, we want to assign probabilities directly to these numbers, as well as to the underlying events. This requires the introduction of some new functions.

(1) **Definition** *Given a sample space* Ω (*with* $\mathcal{F}$ *and* $P(.)$), *a* discrete random variable X *is a function such that for each outcome* ω *in* Ω, $X(\omega)$ *is one of a countable set* D *of real numbers. Formally,* $X(.)$ *is a function with domain* Ω *and range* D, *and so for each* $\omega \in \Omega$, $X(\omega) = x \in D$, *where* D *is a countable (denumerable) subset of the real numbers.* ▲

(2) **Example: Pairs in Poker** How many distinct pairs are there in your poker hand of five cards? Your hand is one outcome ω in the sample space Ω of all possible hands; if you are playing with a full deck, then $|\Omega| = \binom{52}{5}$. The number of pairs X depends on the outcome ω, and obviously $X(\omega) \in \{0, 1, 2\}$, because you can have no more than two pairs. Notice that this holds regardless of how the hand is selected or whether the pack is shuffled. However, this information will be required later to assign probabilities. ●

We always use upper case letters (such as X, Y, T, R, and so on) to denote random variables and lower case letters (x, y, z, etc.) to denote their possible numerical values. You should do the same. Because the possible values of X are countable, we can denote them by $\{x_i; i \in I\}$, where the index set I is a subset of the integers. Very commonly, all the possible values of X are integers, in which case we may denote them simply by x, r, k, j, or any other conventional symbol for integers.

(3) Definition *(a) If X takes only the values 0 or 1, it is called an* indicator *or sometimes a* Bernoulli trial.

(b) If X takes one of only a finite number of values, then it is called simple. ▲

(4) Example Suppose n coins are tossed. Let X_j be the number of heads shown by the jth coin. Then, X_j is obviously zero or one, so we may write $X_j(H) = 1$ and $X_j(T) = 0$.

Let Y be the total number of heads shown by the n coins. Clearly, for each outcome, $\omega \in \Omega$, $Y(\omega) \in \{0, 1, 2, \ldots, n\}$. Thus, X_j is an indicator and Y is simple. ●

It is intuitively clear also that $Y = \sum_1^n X_j$.

We discuss the meaning of this and its implications in Chapter 5.

Finally, note that the sample space Ω need not be countable, even though $X(\omega)$ takes one of a countable set of values.

(5) Example: Darts You throw one dart at a conventional dartboard. A natural sample space Ω is the set of all possible points of impact. This is of course uncountable because it includes every point of the dartboard, much of the wall, and even parts of the floor or ceiling if you are not especially adroit.

However, your score $X(\omega)$ is one of a finite set of integers lying between 0 and 60, inclusive. ●

4.2 Distributions

Next, we need a function, defined on the possible values x of X, to tell us how likely they are. For each such x, there is an event $A_x \subseteq \Omega$, such that

$$\omega \in A_x \Leftrightarrow X(\omega) = x. \tag{1}$$

Hence, just as the probability that any event A in Ω occurs is given by the probability function $\mathbf{P}(A) \in [0, 1]$, the probability that $X(\omega)$ takes any value x is given by a function $\mathbf{P}(A_x) \in [0, 1]$. (We assume that A_x is in $\mathcal{F}$.)

For example, if a coin is tossed and X is the number of heads shown, then $X \in \{0, 1\}$ and $A_1 = H$; $A_0 = T$. Hence, $\mathbf{P}(X = 1) = \mathbf{P}(A_1) = \mathbf{P}(H) = \frac{1}{2}$, if the coin is fair.

This function has its own special name and notation. Given Ω, $\mathcal{F}$, and $\mathbf{P}(.)$:

(2) Definition *A discrete random variable X has a* probability mass function $f_X(x)$ *given by*

$$f_X(x) = \mathbf{P}(A_x).$$

This is also denoted by $\mathbf{P}(X = x)$, which can be thought of as an obvious shorthand for $\mathbf{P}(\{\omega: X(\omega) = x\})$. It is often called the distribution. ▲

For example, let X be the number of pairs in a poker hand, as discussed in Example 4.1.2. If the hand is randomly selected, then

$$f_X(2) = \mathbf{P}(\{\omega: X(\omega) = 2\}) = |\{\omega: X(\omega) = 2\}| \Big/ \binom{52}{5}$$
$$= \binom{13}{2}\binom{4}{2}\binom{4}{2}\binom{44}{1} \Big/ \binom{52}{5} \simeq 0.048.$$

Likewise,

$$f_X(1) = \binom{13}{1}\binom{4}{2}\binom{12}{3}\binom{4}{1}^3 \Big/ \binom{52}{5} \simeq 0.42 ,$$

and hence,

$$f_X(0) = 1 - f_X(1) - f_X(2) \simeq 0.53.$$

Returning to Example 4.1.4 gives an example of great theoretical and historical importance.

(3) Example 4.1.4 Revisited: Binomial Distribution The random variable Y takes the value r, if exactly r heads appear in the n tosses. The probability of this event is $\binom{n}{r}p^r q^{n-r}$, where $p = \mathbf{P}(H) = 1 - q$. Hence, Y has probability mass function

$$f_Y(r) = \binom{n}{r} p^r q^{n-r}, \quad 0 \le r \le n. \qquad \bullet$$

The suffix in $f_X(x)$ or $f_Y(y)$ is included to stress the role of X or Y. Where this is unnecessary or no confusion can arise, we omit it. In the interests of brevity, $f(x)$ is often called simply the *mass function* of X, or even more briefly the p.m.f.

The p.m.f., $f(x) = f_X(x)$, has the following properties: first,

(4)
$$f(x) \ge 0 \quad \text{for } x \in \{x_i: i \in \mathbb{Z}\}$$
$$f(x) = 0 \quad \text{elsewhere.}$$

That is to say, it is positive for a countable number of values of x and zero elsewhere.

Second, if $X(\omega)$ is finite with probability one, then it is called a *proper* random variable and we have

(5)
$$\sum_i f(x_i) = \sum_i \mathbf{P}(A_{x_i}) = \mathbf{P}(\Omega) = 1.$$

Third, we have the *Key Rule*

(6)
$$\mathbf{P}(X \in A) = \sum_{x \in A} f(x).$$

If $\sum_i f(x_i) < 1$, then X is said to be defective or improper. It is occasionally useful to allow X to take values in the extended real line, so that $f_X(\infty)$ has a meaning. In general, it does not.

We remark that any function satisfying (4) and (5) can be regarded as a mass function, in that, given such an $f(.)$, it is quite simple to construct a sample space, probability function, and random variable X, such that $f(x) = \mathbf{P}(X = x)$.

Here are two famous mass functions.

(7) Example: Poisson Distribution Let X be a random variable with mass function

$$f_X(x) = \frac{\lambda^x e^{-\lambda}}{x!}, \quad x \in \{0, 1, 2, \ldots\}, \ \lambda > 0.$$

Then,

$$\sum_{x=0}^{\infty} f(x) = e^{-\lambda} \sum_{x=0}^{\infty} \frac{\lambda^x}{x!} = 1 \quad \text{by Theorem 3.6.9.}$$

Hence, X is proper. This mass function is called the *Poisson distribution* and X is said to be Poisson (or a Poisson random variable), with parameter λ. ●

(8) Example: Negative Binomial Distribution By the negative binomial theorem, for any number q such that $0 < q < 1$, we have

$$(1-q)^{-n} = \sum_{r=0}^{\infty} \binom{n+r-1}{r} q^r.$$

Hence, the function $f(r)$ defined by

$$f(r) = \binom{n+r-1}{r} q^r (1-q)^n, \quad r \geq 0,$$

is a probability mass function. Commonly, we let $1 - q = p$. ●

The following function is also useful; see Figure 4.1 for a simple example

Definition *A discrete random variable X has a* cumulative distribution function *$F_X(x)$, where*

(9)
$$F_X(x) = \sum_{i:x_i \leq x} f(x_i). \qquad \blacktriangle$$

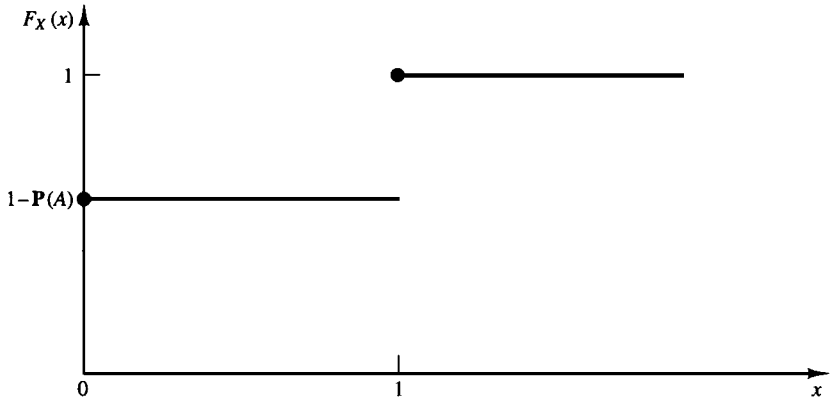


Figure 4.1 The distribution function $F_X(x)$ of the random variable X, which is the indicator of the event A. Thus, the jump at zero is $\mathbf{P}(X = 0) = \mathbf{P}(A^c) = 1 - \mathbf{P}(A)$ and the jump at $x = 1$ is $\mathbf{P}(X = 1) = \mathbf{P}(A)$.

This is also denoted by

$$\mathbf{P}(X \leq x) = \mathbf{P}(\{\omega: X(\omega) \leq x\});$$

it may be referred to simply as the distribution function (or rarely as the c.d.f.), and the suffix X may be omitted. The following properties of $F(x)$ are trivial consequences of the definition (9):

$$F(x) \leq F(y) \quad \text{for } x \leq y. \tag{10}$$

$$1 - F(x) = \mathbf{P}(X > x). \tag{11}$$

$$\mathbf{P}(a < X \leq b) = F(b) - F(a) \quad \text{for } a < b. \tag{12}$$

Some further useful properties are not quite so trivial, in that they depend on Theorem 1.5.2. Thus, if we define the event $B_n = \{X \leq x - 1/n\}$, we find that

$$\begin{aligned}\mathbf{P}(X < x) &= \mathbf{P}\left(\bigcup_{n=1}^{\infty} B_n\right) = \mathbf{P}(\lim_{n\to\infty} B_n) = \lim_{n\to\infty} \mathbf{P}(B_n) \quad \text{by Theorem 1.5.2,} \\ &= \lim_{n\to\infty} F\left(x - \frac{1}{n}\right) = \lim_{y\uparrow x} F(y).\end{aligned}$$

If the random variable X is not defective then, again from (9) (and Theorem 1.5.2), $\lim_{x\to\infty} F(x) = 1$, and $\lim_{x\to-\infty} F(x) = 0$.

The c.d.f. is obtained from the p.m.f. by (9). Conversely, the p.m.f. is obtained from the c.d.f. by

$$f(x) = F(x) - \lim_{y\uparrow x} F(y) \quad \text{where } y < x.$$

When X takes only integer values, this relationship has the following simpler more attractive form: for integer x

$$f(x) = F(x) - F(x - 1). \tag{13}$$

(14) Example: Lottery An urn contains n tickets bearing numbers from 1 to n inclusive. Of these, r are withdrawn at random. Let X be the largest number removed if the tickets are replaced in the urn after each drawing, and let Y be the largest number removed if the drawn tickets are not replaced. Find $f_X(x)$, $F_X(x)$, $f_Y(x)$, and $F_Y(x)$. Show that $F_Y(k) < F_X(k)$, for $0 < k < n$.

Solution The number of ways of choosing r numbers less than or equal to x, with repetition allowed, is x^r. Because there are n^r outcomes,

$$F_X(x) = \left(\frac{x}{n}\right)^r, \quad \text{for } 1 \leq x \leq n,$$

when x is an integer. For any real x, $F_X(x) = ([x]/n)^r$, (where $[x]$ denotes the largest integer which is not greater than x). Hence, for integer x and $1 \leq x \leq n$, by (13),

$$f_X(x) = \left(\frac{x}{n}\right)^r - \left(\frac{x-1}{n}\right)^r;$$

and elsewhere $f_X(x)$ is zero.

Without replacement, the number of ways of choosing r different numbers less than or equal to x is $\binom{x}{r}$. Hence, for integer x, and $1 \leq x \leq n$,

$$F_Y(x) = \binom{x}{r} \Big/ \binom{n}{r}.$$

Hence, again by (13),

$$f_Y(x) = \left(\binom{x}{r} - \binom{x-1}{r}\right) \Big/ \binom{n}{r} = \binom{x-1}{r-1} \Big/ \binom{n}{r},$$

which is of course obvious directly. Furthermore,

$$\begin{aligned} F_Y(k) &= \frac{k!(n-r)!}{(k-r)!n!} < \left(\frac{k}{n}\right)^r \quad \text{for } 1 < k < n \\ &= F_X(k). \end{aligned}$$

●

Because real valued functions of random variables are random variables, they also have probability mass functions.

(15) Theorem *If X and Y are random variables such that $Y = g(X)$, then Y has p.m.f. given by*

$$\sum_{x:g(x)=y} f_X(x).$$

Proof

$$f_Y(y) = \mathbf{P}(g(X) = y) = \sum_{x:g(x)=y} \mathbf{P}(X = x) = \sum_{x:g(x)=y} f_X(x).$$ ■

(16) Example Let X have mass function $f(x)$. Find the mass functions of the following functions of X.

(a) $-X$
(b) $X^+ = \max\{0, X\}$
(c) $X^- = \max\{0, -X\}$
(d) $|X| = X^+ + X^-$
(e) $\operatorname{sgn} X = \begin{cases} \frac{X}{|X|}, & X \neq 0, \\ 0, & X = 0. \end{cases}$

Solution Using Theorem 15 repeatedly, we have:

(a) $f_{-X}(x) = f_X(-x).$

(b) $f_{X^+}(x) = \begin{cases} f_X(x); & x > 0. \\ \sum\limits_{x \leq 0} f_X(x); & x = 0. \end{cases}$

(c) $f_{X^-}(x) = \begin{cases} f_X(-x); & x > 0 \\ \sum\limits_{x \geq 0} f_X(x); & x = 0. \end{cases}$

(d) $$f_{|X|}(x) = \begin{cases} f_X(x) + f_X(-x); & x \neq 0 \\ f_X(0); & x = 0. \end{cases}$$

(e) $$f_{\operatorname{sgn} X}(x) = \begin{cases} \sum_{x>0} f_X(x); & x = 1 \\ f_X(0); & x = 0 \\ \sum_{x<0} f_X(x); & x = -1. \end{cases}$$ ●

Finally, we note that any number m such that $\lim_{x \uparrow m} F(x) \leq \frac{1}{2} \leq F(m)$ is called a *median* of F (or a median of X, if X has distribution F)

4.3 Expectation

(1) **Definition** *Let X be a random variable with probability mass function $f(x)$ such that $\sum_x |x| f(x) < \infty$. The* expected value *of X is then denoted by $\mathbf{E}(X)$ and defined by*

$$\mathbf{E}(X) = \sum_x x f(x).$$

This is also known as the expectation, *or* mean, *or* average *or* first moment *of X.* ▲

Note that $\mathbf{E}(X)$ is the average of the possible values of X weighted by their probabilities. It can thus be seen as a guide to the location of X, and is indeed often called a location parameter. The importance of $\mathbf{E}(X)$ will become progressively more apparent.

(2) **Example** Suppose that X is an indicator random variable, so that $X(\omega) \in \{0, 1\}$. Define the event $A = \{\omega: X(\omega) = 1\}$. Then X is the indicator of the event A; we have $f_X(1) = \mathbf{P}(A)$ and $\mathbf{E}(X) = 0.f_X(0) + 1.f_X(1) = \mathbf{P}(A)$. ●

(3) **Example** Let X have mass function

$$f_X(x) = \frac{4}{x(x+1)(x+2)}, \quad x = 1, 2, \ldots$$

and let Y have mass function

$$f_Y(x) = \frac{1}{x(x+1)}, \quad x = 1, 2, \ldots$$

Show that X does have an expected value and that Y does not have an expected value.

Solution For any $m < \infty$,

$$\sum_{x=1}^{m} |x| f_X(x) = \sum_{x=1}^{m} \frac{4}{(x+1)(x+2)} = 4 \sum_{x=1}^{m} \left(\frac{1}{x+1} - \frac{1}{x+2} \right) = 2 - 4(m+2)^{-1},$$

by successive cancellations in the sum. Hence, the sum converges as $m \to \infty$, and so because $X > 0$, $\mathbf{E}(X) = 2$. However, for the random variable Y,

$$\sum_x |x| f_Y(x) = \sum_{x=1}^{\infty} \frac{1}{x+1},$$

which is not finite. ●

Notice that the condition $\sum |x| f(x) < \infty$ amounts to $\mathbf{E}(X^+) + \mathbf{E}(X^-) < \infty$ (use Example 4.2.16 to see this). A little extension of Definition 1 is sometimes useful. Thus, if $\mathbf{E}(X^-) < \infty$ but $\mathbf{E}(X^+)$ diverges, then we may define $\mathbf{E}(X) = +\infty$. With this extension in Example 3, $\mathbf{E}(Y) = \infty$. Likewise, if $\mathbf{E}(X^+) < \infty$ but $\mathbf{E}(X^-)$ diverges, then $\mathbf{E}(X) = -\infty$. If both $\mathbf{E}(X^+)$ and $\mathbf{E}(X^-)$ diverge, then $\mathbf{E}(X)$ is undefined.

In general, real valued functions of random variables are random variables having a mass function given by Theorem 4.2.15. They may therefore have an expected value. In accordance with Example 3, if $Y = g(X)$, then by definition

$$\mathbf{E}(g(X)) = \sum_i y_i f_Y(y_i).$$

We used this with Example 4.2.16(b) in observing that

$$\mathbf{E}(X^+) = \sum_{x>0} x f_X(x).$$

This was easy because it was easy to find the mass function of X^+ in terms of that of X. It is not such an immediately attractive prospect to calculate (for example) $\mathbf{E}(\cos(\theta X))$ by first finding the mass function of $\cos(\theta X)$. The following theorem is therefore extremely useful.

(4) **Theorem** *Let X be a random variable with mass function $f(x)$, and let $g(.)$ be a real valued function defined on $\mathbb{R}$. Then,*

$$\mathbf{E}(g(X)) = \sum_x g(x) f(x)$$

whenever $\sum_x |g(x)| f(x) < \infty$.

Proof Let (g_j) denote the possible values of $g(X)$, and for each j define the set $A_j = \{x: g(x) = g_j\}$. Then $\mathbf{P}(g(X) = g_j) = \mathbf{P}(X \in A_j)$, and therefore, provided all the following summations converge absolutely, we have

$$\begin{aligned}
\mathbf{E}(g(X)) &= \sum_j g_j \mathbf{P}(g(X) = g_j) = \sum_j g_j \sum_{x \in A_j} f(x) \\
&= \sum_j \sum_{x \in A_j} g(x) f(x), \quad \text{because } g(x) = g_j \quad \text{for } x \in A_j, \\
&= \sum_x g(x) f(x), \quad \text{because } A_j \cap A_k = \phi \quad \text{for } j \neq k.
\end{aligned}$$

■

(5) Example Let X be Poisson with parameter λ. Find $\mathbf{E}(\cos(\theta X))$.

Solution First, recall de Moivre's Theorem that $e^{i\theta} = \cos\theta + i\sin\theta$, where i is an imaginary square root of -1. Now, by Theorem 4,

$$\begin{aligned}\mathbf{E}(\cos(\theta X)) &= \sum_{k=0}^{\infty} \cos(k\theta) e^{-\lambda}\lambda^k / k! \\ &= \mathrm{Re}\left(\sum_{k=0}^{\infty} e^{ik\theta} e^{-\lambda}\lambda^k / k!\right), \quad \text{where Re}(z) \text{ is the real part of } z, \\ &= \mathrm{Re}\left(\exp(\lambda e^{i\theta} - \lambda)\right) \\ &= e^{\lambda(\cos\theta - 1)}\cos(\lambda\sin\theta), \quad \text{using de Moivre's Theorem again.}\end{aligned}$$

●

Now, we can use Theorem 4 to establish some important properties of $\mathbf{E}(.)$.

(6) Theorem *Let X be a random variable with finite mean $\mathbf{E}(X)$, and let a and b be constants. Then:*

(i) $\mathbf{E}(aX + b) = a\mathbf{E}(X) + b$;
(ii) *If* $\mathbf{P}(X = b) = 1$, *then* $\mathbf{E}(X) = b$;
(iii) *If* $\mathbf{P}(a < X \le b) = 1$, *then* $a < \mathbf{E}(X) \le b$;
(iv) *If $g(X)$ and $h(X)$ have finite mean, then*

$$\mathbf{E}(g(X) + h(X)) = \mathbf{E}(g(X)) + \mathbf{E}(h(X)).$$

Proof (i) First, we establish the necessary absolute convergence:

$$\sum_x |ax + b| f(x) \le \sum_x (|a||x| + |b|) f(x) = |a| \sum_x |x| f(x) + |b| < \infty,$$

as required. Hence, by Theorem 4.

$$\mathbf{E}(aX + b) = \sum_x (ax + b) f(x) = a\sum_x x f(x) + b = a\mathbf{E}(X) + b.$$

(ii) Here, X has mass function $f(b) = 1$, so by definition $\mathbf{E}(X) = bf(b) = b$.
(iii) In this case, $f(x) = 0$ for $x \notin (a, b]$, so

$$\mathbf{E}(X) = \sum_x x f(x) \begin{cases} \le \sum_x b f(x) = b; \\ > \sum_x a f(x) = a. \end{cases}$$

(iv) Because $|g(x) + h(x)| \le |g(x)| + |h(x)|$, absolute convergence is quickly established. Hence, by Theorem 4,

$$\begin{aligned}\mathbf{E}(g(X) + h(X)) &= \sum_x (g(x) + h(x)) f(x) = \sum_x g(x) f(x) + \sum_x h(x) f(x) \\ &= \mathbf{E}(g(X)) + \mathbf{E}(h(X)).\end{aligned}$$

■

The following simple corollary is of some importance.

(7) Theorem *If* $\mathbf{E}(X)$ *exists, then* $(\mathbf{E}(X))^2 \le (\mathbf{E}(|X|))^2 \le \mathbf{E}(X^2)$.

Proof First, note that $(|X| - \mathbf{E}(|X|))^2 \ge 0$. Hence, by Theorem 6(iii),

$$\begin{aligned} 0 \le \mathbf{E}((|X| - \mathbf{E}(|X|))^2 &= \mathbf{E}(|X|^2) - (\mathbf{E}(|X|))^2, \quad \text{by Theorem 6(iv) and 6(ii)} \\ &= \mathbf{E}(X^2) - (\mathbf{E}(|X|))^2, \end{aligned}$$

which proves the second inequality. Also, $|X| - X \ge 0$, so by 6(iv) and 6(iii)

$$\mathbf{E}(X) \le \mathbf{E}(|X|),$$

which proves the first inequality. ■

(8) Example: Uniform Distribution Recall that an urn contains n tickets numbered from 1 to n. You take one ticket at random; it bears the number X. Find $\mathbf{E}(X)$ and $\mathbf{E}(X^2)$, and verify that Theorem 7 holds explicitly.

Solution The mass function of X is $\mathbf{P}(X = k) = 1/n$. (Because it distributes probability evenly over the values of X, it is called the *uniform distribution.*) Hence,

$$\begin{aligned} \mathbf{E}(X) = \sum_{x=1}^{n} \frac{x}{n} &= \frac{1}{n} \sum_{x=1}^{n} \frac{1}{2}(x(x+1) - x(x-1)) \\ &= \frac{1}{2}(n+1) \quad \text{by successive cancellation.} \end{aligned}$$

Likewise, using Theorems 4 and 6(iv),

$$\begin{aligned} \mathbf{E}(X^2) + \mathbf{E}(X) = \sum_{x=1}^{n} \frac{x^2 + x}{n} &= \frac{1}{n} \sum_{x=1}^{n} \frac{1}{3}(x(x+1)(x+2) - (x-1)x(x+1)) \\ &= \frac{1}{3}(n+1)(n+2) \quad \text{by successive cancellation.} \end{aligned}$$

Hence,

$$\mathbf{E}(X^2) = \frac{1}{6}(n+1)(2n+1) \ge \frac{1}{4}(n+1)^2 = (\mathbf{E}(X))^2. \qquad \bullet$$

In practice, we are often interested in the expectations of two particularly important collections of functions of X; namely, $(X^k; k \ge 1)$ and $([X - \mathbf{E}(X)]^k; k \ge 1)$.

(9) Definition *Let* X *have mass function* $f(x)$ *such that* $\sum_x |x|^k f(x) < \infty$. *Then,*

(a) *The* k*th* moment *of* X *is* $\mu_k = \mathbf{E}(X^k)$.
(b) *The* k*th* central moment *of* X *is* $\sigma_k = \mathbf{E}((X - \mathbf{E}(X))^k)$.
(c) *The* k*th* factorial moment *of* X *is* $\mu^{(k)} = \mathbf{E}(X(X-1)\dots(X-k+1))$.

In particular, σ_2 *is called the* variance *of* X *and is denoted by* σ^2, σ_X^2, *or* var (X). *Thus,*

$$\text{var}\,(X) = \mathbf{E}((X - \mathbf{E}(X))^2). \qquad \blacktriangle$$

Example: Indicators Let X be the indicator of the event A (recall Example 4.3.2). Because $X^k(\omega) = X(\omega)$ for all k and ω, we have

$$\mu_k = \mathbf{E}(X^k) = \mathbf{P}(A).$$

Also, $\text{var}\,(X) = \mathbf{P}(A)\mathbf{P}(A^c)$, and

$$\mu^{(k)} = \begin{cases} \mathbf{P}(A); & k = 1 \\ 0; & k > 1. \end{cases}$$

●

(10) Example Show that if $\mathbf{E}(X^2) < \infty$, and a and b are constants then

$$\text{var}\,(aX + b) = a^2 \,\text{var}\,(X).$$

Solution Using Theorem 6(i) and the definition of variance,

$$\text{var}\,(aX + b) = \mathbf{E}((a(X - \mathbf{E}(X)) + b - b)^2) = \mathbf{E}(a^2(X - \mathbf{E}(X))^2) = a^2 \,\text{var}\,(X).$$

●

Sometimes the tail of a distribution, $\mathbf{P}(X > x)$, has a simpler form than the mass function $f(x)$. In these and other circumstances, the following theorems are useful.

(11) Theorem *If $X \geq 0$ and X takes integer values, then* $\mathbf{E}(X) = \sum_{x=1}^{\infty} \mathbf{P}(X \geq x)$.

Proof By definition,

$$\mathbf{E}(X) = \sum_{x=1}^{\infty} x f(x) = \sum_{x=1}^{\infty} f(x) \sum_{r=1}^{x} 1.$$

Because all terms are nonnegative, we may interchange the order of summation to obtain

$$\sum_{x=1}^{\infty} \sum_{r=x}^{\infty} f(r) = \sum_{x=1}^{\infty} \mathbf{P}(X \geq x).$$

■

This *tail-sum theorem* has various generalizations; we state one.

(12) Theorem *If $X \geq 0$ and $k \geq 2$, then*

$$\mu^{(k)} = \mathbf{E}(X(X - 1) \ldots (X - k + 1)) = k \sum_{x=k}^{\infty} (x - 1) \ldots (x - k + 1)\mathbf{P}(X \geq x).$$

Proof This is proved in the same way as Theorem 11, by changing the order of summation on the right-hand side. ■

(13) Example: Waiting–The Geometric Distribution A biased coin shows a head with probability p or a tail with probability $q = 1 - p$. How many times do you expect to toss the coin until it first shows a head? Find the various second moments of this waiting time.

Solution Let the required number of tosses until the first head be T. Then because they are independent, $\mathbf{P}(T = x) = q^{x-1}p; x \geq 1$. ($T$ is said to have the *geometric* distribution.)

Hence,

$$\begin{aligned}\mathbf{E}(T) &= \sum_{x=1}^{\infty} xq^{x-1}p = \frac{p}{(1-q)^2}, \quad \text{by (3.6.12) with } n = 2, \\ &= \frac{1}{p}.\end{aligned}$$

Alternatively, we can use Theorem 11 as follows. Using the independence of tosses again gives $\mathbf{P}(T > x) = q^x$, so

$$\mathbf{E}(T) = \sum_{x=0}^{\infty} \mathbf{P}(T > x) = \sum_{x=0}^{\infty} q^x = \frac{1}{p}.$$

For the second factorial moment, by Theorem 12

$$\mu^{(2)} = 2\sum_{x=2}^{\infty}(x-1)q^{x-1} = \frac{2q}{p^2}, \quad \text{by (3.6.12) again.}$$

Hence, the second moment is

$$\mathbf{E}(T^2) = \mathbf{E}(T(T-1)) + \mathbf{E}(T) = \frac{2q}{p^2} + \frac{1}{p} = \frac{1+q}{p^2}.$$

Finally, the second central moment is

$$\sigma_2 = \mathbf{E}((T - \mathbf{E}(T))^2) = \mathbf{E}(T^2) - (\mathbf{E}(T))^2 = \frac{1+q}{p^2} - \frac{1}{p^2} = \frac{q}{p^2}. \quad \bullet$$

(14) Example Let X have mass function

$$f_X(x) = \frac{a}{x^2}; \quad x = 1, 2, 3, \ldots$$

and Y have mass function

$$f_Y(y) = \frac{b}{y^2}; \quad y = \pm 1, \pm 2, \ldots$$

(a) Find a and b.
(b) What can you say about $\mathbf{E}(X)$ and $\mathbf{E}(Y)$?

Solution (a) Because $f_X(x)$ is a mass function

$$1 = \sum_x f_X(x) = a\sum_{x=1}^{\infty} \frac{1}{x^2} = a\frac{\pi^2}{6}.$$

Hence, $a = 6\pi^{-2}$. Likewise, $b = 3\pi^{-2}$.

(b) We have

$$\mathbf{E}(X) = a\sum_{x=1}^{\infty} xf_X(x) = a\sum_{x=1}^{\infty} \frac{1}{x} = \infty.$$

Because $\mathbf{E}(Y^+)$ and $\mathbf{E}(Y^-)$ both diverge, $\mathbf{E}(Y)$ does not exist. $\bullet$

(15) Example: Coupons Each packet of an injurious product is equally likely to contain any one of n different types of coupon, independently of every other packet. What is the expected number of packets you must buy to obtain at least one of each type of coupon?

Solution Let A_n^r be the event that the first r coupons you obtain do not include a full set of n coupons. Let C_k^r be the event that you have not obtained one of the kth coupon in the first r. Then,

$$A_n^r = \bigcup_{k=1}^{n} C_k^r.$$

We may calculate:

$$\mathbf{P}(C_1^r) = \left(\frac{n-1}{n}\right)^r,$$
$$\mathbf{P}(C_i^r \cap C_j^r) = \left(\frac{n-2}{n}\right)^r; \quad i \neq j,$$

and, in general, for any set S_j of j distinct coupons

$$\mathbf{P}\left(\bigcap_{i \in S_j} C_i^r\right) = \left(\frac{n-j}{n}\right)^r.$$

Hence, by (1.4.8),

$$\mathbf{P}(A_n^r) = \mathbf{P}\left(\bigcup_{k=1}^{n} C_k^r\right) = \sum_{j=1}^{n} (-1)^{j+1} \binom{n}{j} \left(1 - \frac{j}{n}\right)^r,$$

because for each j there are $\binom{n}{j}$ sets S_j.

Now, let R be the number of packets required to complete a set of n distinct coupons. Because A_n^r occurs if and only if $R > r$, we have $\mathbf{P}(R > r) = \mathbf{P}(A_n^r)$. Hence, by Theorem 11,

$$\begin{aligned}\mathbf{E}(R) &= \sum_{r=0}^{\infty} \mathbf{P}(R > r) = \sum_{r=0}^{\infty} \sum_{j=1}^{n} (-1)^{j+1} \binom{n}{j} \left(1 - \frac{j}{n}\right)^r \\ &= \sum_{j=1}^{n} (-1)^{j+1} \binom{n}{j} \frac{n}{j} = n u_n, \quad \text{say.}\end{aligned}$$

Now,

$$\begin{aligned}u_{n+1} - u_n &= \sum_{j=1}^{n+1} (-1)^{j+1} \binom{n+1}{j} \frac{1}{j} - \sum_{j=1}^{n} (-1)^{j+1} \binom{n}{j} \frac{1}{j} \\ &= \frac{(-1)^{n+2}}{n+1} + \sum_{j=1}^{n} \frac{(-1)^{j+1}}{j} \left(\binom{n+1}{j} - \binom{n}{j} \right) \\ &= \frac{(-1)^{n+2}}{n+1} + \sum_{j=1}^{n} \frac{(-1)^{j+1}}{n+1} \binom{n+1}{j} \\ &= \frac{1}{n+1},\end{aligned} \tag{16}$$

because

$$\sum_{j=0}^{n+1}(-1)^{j+1}\binom{n+1}{j} = (1-1)^{n+1} = 0.$$

Hence, iterating (16),

$$u_n = \sum_{j=1}^{n}\frac{1}{n-j+1},$$

so that

$$\mathbf{E}(R) = \sum_{j=1}^{n}\frac{n}{n-j+1}.$$

In Chapter 5, we discover a much easier method of obtaining this result. ●

4.4 Conditional Distributions

Let Ω be some sample space, X some random variable defined on Ω, and $\mathbf{P}(.)$ a probability function defined on Ω. Now suppose that we are given that some event $B \subseteq \Omega$ occurs, with $\mathbf{P}(B) > 0$. Just as we argued in Chapter 2 that this gives rise to a conditional probability function, so we now conclude that this gives rise to a conditional distribution of X given B.

In fact, using (4.2.1), we write

(1) $$\mathbf{P}(X(\omega) = x|B) = \mathbf{P}(A_x|B) = \mathbf{P}(A_x \cap B)/\mathbf{P}(B) \geq 0,$$

where as usual $A_x = \{\omega: X(\omega) = x\}$. Furthermore, because $\bigcup_x A_x = \Omega$ and $A_x \cap A_y = \phi$, whenever $x \neq y$ we have

(2) $$\sum_x \mathbf{P}(A_x|B) = \sum_x \mathbf{P}(A_x \cap B)/\mathbf{P}(B) = \mathbf{P}(\Omega \cap B)/\mathbf{P}(B) = 1.$$

Hence, the function $f(x|B)$ defined by

(3) $$f(x|B) = \mathbf{P}(A_x|B) = \mathbf{P}(X = x|B)$$

is a probability mass function, in that $f(x|B) \geq 0$ and $\sum_x f(x|B) = 1$. It is the *conditional mass function* of X given B.

(4) Example Let X be uniformly distributed on $\{1, 2, \ldots, n\}$, and let B be the event that $a \leq X \leq b$, where $1 \leq a < b \leq n$. Find the mass function of X given B.

Solution Obviously,

$$\mathbf{P}(B) = \sum_{i=a}^{b}\frac{1}{n} = (b-a+1)/n$$

and

$$\mathbf{P}(\{X = k\} \cap B) = \begin{cases} \dfrac{1}{n}; & a \le k \le b \\ 0; & \text{otherwise.} \end{cases}$$

Hence,

$$f(x|B) = \begin{cases} \dfrac{1}{b-a+1}; & a \le x \le b \\ 0; & \text{otherwise} \end{cases}$$

Thus, given that X lies in B, it is uniformly distributed over B. ●

Because $f(x|B)$ is a probability mass function, it may have an expectation. In line with Definition 4.3.1, we require that $\sum_x |x| f(x|B) < \infty$. If this condition is satisfied, then the *conditional expectation* of X given B is denoted by $\mathbf{E}(X|B)$ and defined by

(5) $$\mathbf{E}(X|B) = \sum_x x f(x|B).$$

Expectation and conditional expectation are related by the following exceptionally important result.

(6) Theorem *Let X be a random variable with mean $\mathbf{E}(X)$, and let B be an event such that $\mathbf{P}(B)\mathbf{P}(B^c) > 0$. Then,*

$$\mathbf{E}(X) = \mathbf{E}(X|B)\mathbf{P}(B) + \mathbf{E}(X|B^c)\mathbf{P}(B^c).$$

Proof By conditional probability,

$$\begin{aligned} f(x) = \mathbf{P}(X = x) &= \mathbf{P}(\{X = x\} \cap B) + \mathbf{P}(\{X = x\} \cap B^c) \\ &= f(x|B)\mathbf{P}(B) + f(x|B^c)\mathbf{P}(B^c). \end{aligned}$$

Hence,

$$\mathbf{E}(X) = \sum_x x f(x) = \mathbf{P}(B) \sum_x x f(x|B) + \mathbf{P}(B^c) \sum_x x f(x|B^c),$$

as required. ■

More generally, it is shown in exactly the same way that if $(B_i; i \ge 1)$ is a collection of events such that

(i) $\bigcup_i B_i = \Omega$,
(ii) $B_i \cap B_j = \phi; \quad i \ne j$, and
(iii) $\mathbf{P}(B_i) > 0$,

then

$$\mathbf{E}(X) = \sum_i \mathbf{E}(X|B_i)\mathbf{P}(B_i) \tag{7}$$

whenever the summation is absolutely convergent.

Finally, we make the small but useful observation that if $A \subseteq B$, then

$$\mathbf{E}(X|A \cap B) = \mathbf{E}(X|A). \tag{8}$$

(9) Example A coin is tossed repeatedly. As usual, for each toss, $\mathbf{P}(H) = p = 1 - q = 1 - \mathbf{P}(H^c)$. The outcome is a sequence of runs of heads alternating with runs of tails; the first run can be of either heads or tails. Let the length of the nth run be R_n. For all k and j, show that $\mathbf{E}(R_{2k+1}) \geq \mathbf{E}(R_{2j})$ and $\text{var}\,(R_{2k+1}) \geq \text{var}\,(R_{2j})$, with equality in each case if and only if $p = q = \frac{1}{2}$.

Solution Let X be the number of heads shown before the first appearance of a tail. We know that

$$\mathbf{P}(X = k) = p^k q, \quad k \geq 0. \tag{10}$$

Let us consider the mass function of X conditional on the first toss. Given that the first toss is H, let X' be the further number of tosses before the first tail. By independence,

$$\mathbf{P}(X' = k) = p^k q = \mathbf{P}(X = k).$$

Hence, conditional on H, we have $X = 1 + X'$, and conditional on H^c, we have $X = 0$. Therefore, by Theorem 6,

$$\mathbf{E}(X) = p\mathbf{E}(X|H) + q\mathbf{E}(X|H^c) = p(1 + \mathbf{E}(X')) + 0 = p + p\mathbf{E}(X) \tag{11}$$
$$\text{because } \mathbf{E}(X) = \mathbf{E}(X').$$

Thus, $\mathbf{E}(X) = p/q$, which of course we could have obtained directly from (10); we chose to do it this way to display the new technique. Likewise, if Y is the number of tails before the first head, $\mathbf{E}(Y) = q/p$.

Now R_{2k+1} is a run of heads if and only if the first toss is a head. Hence, again using independence,

$$\begin{aligned}\mathbf{E}(R_{2k+1}) &= \mathbf{E}(R_{2k+1}|H)p + \mathbf{E}(R_{2k+1}|H^c)q = \mathbf{E}(1 + X')p + \mathbf{E}(1 + Y')q \\ &= \left(1 + \frac{p}{q}\right)p + \left(1 + \frac{q}{p}\right)q = \frac{p}{q} + \frac{q}{p}.\end{aligned}$$

Likewise, R_{2k} is a run of heads if and only if the first toss yields a tail, so

$$\mathbf{E}(R_{2k}) = \left(1 + \frac{p}{q}\right)q + \left(1 + \frac{q}{p}\right)p = 2 \leq \frac{p}{q} + \frac{q}{p},$$

with equality if and only if $p = \frac{1}{2} = q$. [Because $(p - q)^2 > 0$, for $p \neq q$.] Now, var $(R_{2k}) = \mathbf{E}(R_{2k}^2) - 4$. Arguing as above, and using conditional probability again, yields

$$\mathbf{E}(R_{2k}^2) = q\mathbf{E}((1 + X')^2) + p\mathbf{E}((1 + Y')^2) = q\left(\frac{1 + p}{q^2}\right) + p\left(\frac{1 + q}{p^2}\right)$$

and so

$$\text{var}\,(R_{2k}) = \frac{1 + (p-q)^2 - 2pq}{pq}. \tag{12}$$

Likewise,

$$\mathbf{E}(R_{2k+1}^2) = q\left(\frac{1+q}{p^2}\right) + p\left(\frac{1+p}{q^2}\right)$$

and so

$$\text{var}\,(R_{2k+1}) = \frac{q}{p^2} + \frac{p}{q^2} - 2. \tag{13}$$

Now

$$\text{var}\,(R_{2k+1}) - \text{var}\,(R_{2k}) = \frac{p^4 + q^4 + 2p^2q^2 - pq}{p^2q^2} = \frac{(p^3 - q^3)(p-q)}{p^2q^2} \geq 0$$

with equality if and only if $p = q = \frac{1}{2}$. ●

4.5 Sequences of Distributions

If an experiment is repeated indefinitely, it may give rise to a sequence $(F_n(x); n \geq 1)$ of distributions.

(1) Definition *Let $f(x)$ be a probability mass function that is nonzero for $x \in D$, and zero for $x \in \mathbb{R} \backslash D = C$. Let $F(x)$ be the corresponding distribution function*

$$F(x) = \sum_{x_i \leq x} f(x_i).$$

A sequence of distribution functions $F_n(x)$ is said to converge to $F(x)$ if, as $n \to \infty$,

$$F_n(x) \to F(x) \qquad \text{for } x \in C. \quad ▲$$

One special case is important to us; if D is included in the integers, then $F_n(x)$ converges to $F(x)$ if, for all x, $f_n(x) \to f(x)$ as $n \to \infty$.

(2) Example: Matching revisited In Example 3.18, we showed that the probability of exactly r matches in n random assignments of letters is

$$p(n, r) = \frac{1}{r!} \sum_{k=0}^{n-r} \frac{(-)^k}{k!} \to \frac{e^{-1}}{r!}$$

as $n \to \infty$. This shows that as $n \to \infty$ the number of matches has a Poisson distribution (with parameter 1) in the limit. ●

(3) Example: Ménages Revisited In Problem 3.38, we found the probability that exactly m couples were adjacent when seated randomly at a circular table (alternating the

sexes) is

$$p_m = \frac{2}{m!}\sum_{k=0}^{n-m}\frac{(-)^k(n-m-k)!(2n-m-k-1)!}{k!(2n-2m-2k)!n!} \to \frac{2^m}{m!}\sum_{k=0}^{\infty}(-)^k\frac{2^k}{k!} = \frac{2^m e^{-2}}{m!}$$

as $n \to \infty$. Thus, the number of adjacent couples is Poisson with parameter 2 in the limit as $n \to \infty$. ●

Finally, we note that the appearance of the Poisson distribution in Examples 2 and 3 is significant. This distribution commonly arises in limits of this type, and that is one of the reasons for its major importance.

4.6 Inequalities

Calculating the exact probability that X lies in some set of interest is not always easy. However, simple bounds on these probabilities will often be sufficient for the task in hand. We start with a basic inequality.

(1) Theorem: Basic Inequality *If $h(x)$ is a nonnegative function, then, for $a > 0$*

$$\mathbf{P}(h(X) \geq a) \leq \mathbf{E}(h(X))/a).$$

Proof Define the following function of X:

$$I(h \geq a) = \begin{cases} 1 & \text{whenever } h(X) \geq a \\ 0 & \text{otherwise} \end{cases}$$

Observe that I is an indicator, and so by Example 4.3.2, $\mathbf{E}(I) = \mathbf{P}(h(X) \geq a)$. Now, by its construction I satisfies $h(X) - aI \geq 0$, and so by Theorem 4.3.6 [parts (iii) and (iv)],

$$\mathbf{E}(h(X)) \geq a\mathbf{E}(I) = a\mathbf{P}(h(X) \geq a).$$ ■

The following useful inequalities can all be proved using Theorem 1 or by essentially the same method. You should do some as exercises. For any $a > 0$, we have:

Markov's Inequality

$$\mathbf{P}(|X| \geq a) \leq \mathbf{E}(|X|)/a. \tag{2}$$

Chebyshov's Inequality[†]

$$\mathbf{P}(|X| \geq a) \leq \mathbf{E}(X^2)/a^2. \tag{3}$$

[†] Some writers use the transliteration "Chebyshev". They then have to remember that the second "e" is pronounced as "o".

One-Sided Chebyshov's Inequality

$$\mathbf{P}(X - \mathbf{E}(X) \geq a) \leq \frac{\operatorname{var}(X)}{a^2 + \operatorname{var}(X)}. \tag{4}$$

Generalized Markov Inequality If $h(x)$ is increasing for $x > 0$, even, and nonnegative then

$$\mathbf{P}(|X| \geq a) \leq \mathbf{E}(h(X))/h(a). \tag{5}$$

If X is nonnegative, then

$$\mathbf{P}(X > a) \leq \frac{\mathbf{E}(X)}{a}. \tag{6}$$

If $c > 0$, then

$$\mathbf{P}(X > a) \leq \frac{\mathbf{E}((X + c)^2)}{(a + c)^2}, \tag{7}$$

and

$$\mathbf{P}(X > a) \leq \mathbf{E}(\exp(c(X - a))). \tag{8}$$

Here is one important application.

(9) Example Let X be a random variable such that $\operatorname{var}(X) = 0$. Show that X is constant with probability one.

Solution By (3), for any integer $n \geq 1$,

$$\mathbf{P}\left(|X - \mathbf{E}(X)| > \frac{1}{n}\right) \leq n^2 \operatorname{var}(X) = 0. \tag{10}$$

Hence, defining the events $C_n = \{|X - \mathbf{E}(X)| > 1/n\}$, we have

$$\begin{aligned}\mathbf{P}(X \neq \mathbf{E}(X)) &= \mathbf{P}\left(\bigcup_{n=1}^{\infty} C_n\right) = \mathbf{P}(\lim_{n\to\infty} C_n) = \lim_{n\to\infty} \mathbf{P}(C_n) \qquad \text{by Theorem 1.5.2} \\ &= 0.\end{aligned}$$

●

An important concept that crops up in many areas of pure and applied mathematics is that of *convexity*. We are interested in the following manifestation of this.

(11) Definition *A function $g(x)$ (from $\mathbb{R}$ to $\mathbb{R}$) is called* convex *if, for all a, there exists $\lambda(a)$ such that*

$$g(x) \geq g(a) + \lambda(a)(x - a), \quad \textit{for all } x. \tag{12}$$

▲

If $g(x)$ is differentiable, then a suitable λ is given by $\lambda(a) = g'(a)$ and (12) takes the form

$$g(x) \geq g(a) + g'(a)(x - a). \tag{13}$$

This says that a convex function lies above all its tangents. If g is not differentiable, then there may be many choices for λ; draw a picture of $g(x) = |x|$ at $x = 0$ to see this. (There are several other definitions of a convex function, all equivalent to Definition 11.) We are interested in the following property of convex functions.

(14) Theorem: Jensen's Inequality *Let X be a random variable with finite mean and $g(x)$ a convex function. Then,*

$$\mathbf{E}(g(X)) \geq g(\mathbf{E}(X)). \tag{15}$$

Proof Choosing $a = \mathbf{E}(X)$ in (12), we have

$$g(X) \geq g(\mathbf{E}(X)) + \lambda(X - \mathbf{E}(X)).$$

Taking the expected value of each side gives (15). ■

For example, $g(x) = |x|$ and $g(x) = x^2$ are both convex, so $\mathbf{E}(|X|) \geq |\mathbf{E}(X)|$ and $\mathbf{E}(X^2) \geq (\mathbf{E}(X))^2$. This is Theorem 4.3.7. Here is a less trivial example.

(16) Example Let X be a positive random variable. Show that $\mathbf{E}(\log X) \leq \log \mathbf{E}(X)$.

Solution This follows immediately from Jensen's inequality if we can show that $-\log x$ is convex. Fortunately, this is easy, as follows. By definition, for $x > 0$,

$$\begin{aligned} -\log x &= \int_x^1 y^{-1} dy = \int_a^1 y^{-1} dy + \int_x^a y^{-1} dy, \quad \text{for } a > 0, \\ &= -\log a + \int_x^a y^{-1} dy \geq -\log a + \int_x^a a^{-1} dy = -\log a - \frac{1}{a}(x - a), \end{aligned}$$

and this is (12) with $\lambda(a) = -a^{-1}$. ●

Example 16 has many important applications, of which we see more later. Here is one to begin with.

(17) Example: Arithmetic–Geometric Means Inequality Let $(x_i; 1 \leq i \leq n)$ be any collection of positive numbers and $(p_i; 1 \leq i \leq n)$ any collection of positive numbers such that $\sum_i p_i = 1$. Show that

$$p_1 x_1 + p_2 x_2 + \cdots + p_n x_n \geq {x_1}^{p_1} {x_2}^{p_2} \cdots {x_n}^{p_n}. \tag{18}$$

Solution Let X be the random variable with probability mass function $\mathbf{P}(X = x_i) = p_i; 1 \leq i \leq n$. Then, from (16),

$$\begin{aligned} \log \mathbf{E}(X) &= \log(p_1 x_1 + \cdots + p_n x_n) \geq \mathbf{E}(\log X) \\ &= p_1 \log x_1 + \cdots + p_n \log x_n = \log({x_1}^{p_1} {x_2}^{p_2} \cdots {x_n}^{p_n}). \end{aligned}$$

The result (18) follows because $\log x$ is an increasing function. ●

(19) Example: AM/GM Inequality In the special case when $p_i = 1/n$, $1 \le i \le n$, then (18) takes the form

$$\frac{1}{n}\sum_1^n x_i \ge \left(\prod_{i=1}^n x_i\right)^{1/n}. \qquad \bullet$$

4.7 Review and Checklist for Chapter 4

We have seen that many experiments have numerical outcomes; and when they do not, the sample space can be usefully mapped to suitable points on the line. Such real-valued functions defined on the sample space are called random variables. This chapter defined some important types of random variable, and considered several important named examples. We introduced the key concepts of distribution, expectation, functions of random variables, conditional distributions, and sequences of distributions. In Table 4.1 we display the distribution, expectation, and variance for some important specific random variables.

SYNOPSIS OF FORMULAE: In general, every random variable has a distribution function

$$F_X(x) = \mathbf{P}(X \le x).$$

Discrete random variables, taking one of a countable set of possible values, have a probability mass function

$$f_X(x) = \mathbf{P}(X = x).$$

When X is integer valued, we can write

$$f_X(x) = F(x) - F(x-1).$$

Table 4.1. *Discrete random variables*

Distribution	$f(x)$	Mean	Variance
Uniform	n^{-1}, $1 \le x \le n$	$\frac{1}{2}(n+1)$	$\frac{1}{12}(n^2-1)$
Bernoulli	$p^x(1-p)^{1-x}$, $x \in \{0,1\}$	p	$p(1-p)$
Binomial	$\binom{n}{x} p^x(1-p)^{n-x}$, $0 \le x \le n$	np	$np(1-p)$
Geometric	$(1-p)^{x-1}p$, $x \ge 1$	p^{-1}	$(1-p)p^{-2}$
Poisson	$\frac{e^{-\lambda}\lambda^x}{x!}$, $x \ge 1$	λ	λ
Negative binomial	$\binom{x-1}{k-1} p^k(1-p)^{x-k}$, $x \ge k$	kp^{-1}	$k(1-p)p^{-2}$
Hypergeometric	$\frac{\binom{m}{x}\binom{w}{n-x}}{\binom{m+w}{n}}$, $0 \le x \le n$	$\frac{nm}{m+w}$	$\frac{nmw(m+w-n)}{(m+w-1)(m+w)^2}$

In any case, we have the

Key Rule:

$$\mathbf{P}(X \in A) = \sum_{x \in A} f_X(x).$$

Expectation: A discrete random variable has an expected value if $\sum_x |x| f_X(x) < \infty$. In this case,

$$\mathbf{E}X = \sum_x x f_x(x).$$

When X is integer valued and nonnegative,

$$\mathbf{E}X = \sum_{x=0}^{\infty} \mathbf{P}(X > x) = \sum_{x=0}^{\infty} (1 - F(x)).$$

Functions: Suppose that discrete random variables X and Y are such that $Y = g(X)$ for some function $g(.)$. Then,

$$f_Y(y) = \sum_{x:g(x)=y} f_X(x).$$

Also,

$$\mathbf{E}Y = \sum_x g(x) f_X(x)$$

[provided that $\sum_x |g(x)| f_X(x) < \infty$.]
It follows that in any case where each side exists:

$$\mathbf{E}[ag(X) + bh(X)] = a\mathbf{E}g(X) + b\mathbf{E}h(X).$$

Variance: For any random variable X, the variance σ^2 is given by

$$\text{var } X = \mathbf{E}(X - \mathbf{E}X)^2 = \mathbf{E}X^2 - (\mathbf{E}X)^2 = \sigma^2 \geq 0.$$

The number $\sigma \geq 0$ is called the *standard deviation*, and in particular we have, for constants a and b,

$$\text{var}\,(aX + b) = a^2 \text{var} X.$$

Moments: The kth moment of X is

$$\mu_k = \mathbf{E}X^k, \quad k \geq 1;$$

usually we write $\mu_1 = \mu$. The kth central moment of X is

$$\sigma_k = \mathbf{E}(X - \mathbf{E}X)^k, \quad k \geq 1;$$

usually we write $\sigma_2 = \sigma^2$.

Conditioning: Any event B in Ω may condition a random variable X, leading to a conditional distribution function

$$F_{X|B}(x|B) = \mathbf{P}(X \leq x|B)$$

and a conditional mass function

$$f_{X|B}(x|B) = \mathbf{P}(X = x|B);$$

Key Rule:

$$\mathbf{P}(X \in A|B) = \sum_{x \in A} f_{X|B}(x|B).$$

This distribution may have an expected value, called
Conditional expectation:

$$\mathbf{E}(X|B) = \sum_{x} x f_{X|B}(x|B).$$

In particular,

$$\mathbf{E}X = \mathbf{E}(X|B)\mathbf{P}(B) + \mathbf{E}(X|B^c)\mathbf{P}(B^c)$$

and if $(B_i; i \geq 1)$ is a partition of Ω

$$\mathbf{E}X = \sum_{i} \mathbf{E}(X|B_i)\mathbf{P}(B_i).$$

Similarly, a random variable X may condition an event B, so that

$$\mathbf{P}(B|X = x) = \mathbf{P}(B \cap \{X = x\})/\mathbf{P}(X = x),$$

yielding

$$\mathbf{P}(B) = \sum_{x} \mathbf{P}(B|X = x) f_X(x).$$

Checklist of Terms for Chapter 4

4.1 discrete random variable
Bernoulli trial
indicator
simple random variable
4.2 probability distribution
probability mass function
binomial distribution
Poisson distribution
negative binomial distribution
distribution function
4.3 expectation and expected value
uniform distribution
moments
variance
tail sum
geometric distribution
4.4 conditional mass function
conditional expectation
4.5 sequences of distributions

4.6 Markov's inequality
Chebyshov's inequality
Jensen's inequality
Arithmetic-geometric means inequality

WORKED EXAMPLES AND EXERCISES

4.8 Example: Royal Oak Lottery

This eighteenth-century lottery paid winners 28 to 1; the chance of winning at any given bet was 2^{-5}, independently of other bets. Gamesters (as usual) complained that the odds were unfair. It is reported by de Moivre (in *Doctrine of Chances*, 1756) that the Master of the Ball maintained that any particular point of the Ball should come up once in 22 throws; he offered to bet on this (at evens) at any time, and did so when required. The seeming contradiction between the 2^{-5} chance at any bet, with 22 throws for any chance to come up, so perplexed the gamesters that they began to think they had the advantage; so they played on and continued to lose.

Explain why there is no contradiction.

Solution Let $\mathbf{P}$ be a point of the Ball. Let T be the number of trials required to yield $\mathbf{P}$ for the first time. At each trial, $\mathbf{P}$ fails to appear with probability 31/32, and $T > k$ if and only if the first k trials do not yield $\mathbf{P}$. Hence, by independence,

$$\mathbf{P}(T > k) = \left(\frac{31}{32}\right)^k.$$

Now,

$$\left(\frac{31}{32}\right)^{22} \simeq 0.49 < 0.5.$$

Hence,

$$\mathbf{P}(T \leq 22) > 0.5.$$

[However, note that $\mathbf{E}(T) = \sum_0^\infty \mathbf{P}(T > k) = (1 - \frac{31}{32})^{-1} = 32$.]

Thus, by betting on the event $T \leq 22$, the Master of the Ball was giving himself a better than evens chance of winning. However, if we let W be the profit to the gambler of a \$1 stake wagered on $\mathbf{P}$ turning up, we have $\mathbf{P}(W = 28) = \frac{1}{32}$ and $\mathbf{P}(W = -1) = \frac{31}{32}$. Hence, $\mathbf{E}(W) = \frac{28}{32} - \frac{31}{32} = -\frac{3}{32}$. A *loss*.

Thus, in the long run, the gambler will surely lose at a rate of nearly 10% of his stake each play. (See Example 4.18 for a proof of this.)

Remark The Master of the Ball was exploiting the fact that the median of the distribution of T is less than its mean. See Problem 4.51 for bounds on this difference.

Note that T has a *geometric* distribution.

(1) Exercise Give an example of a distribution for which the median is larger than the mean.
(2) Exercise Find: (a) var (T) and (b) $\mu_T^{(k)}$.
(3) Exercise Which of the following strategies gives the gambler a better chance of winning if she takes up the offer of a bet on **P** not occurring in 22 trials:
(a) Making such a bet immediately?
(b) Waiting for a run of 22 trials during which **P** has not appeared?
(c) Waiting until **P** has appeared in consecutive trials and then betting on its nonappearance in the following 22?
(4) Exercise Calculate $\mathbf{P}(T > j + k | T > j)$. Explain the significance of your answer.

4.9 Example: Misprints

Each printed character in a book is misprinted independently with probability p, or is correct with probability $1 - p$. Let n be the number of characters in the book, and let X be the number of misprinted characters.

(a) Find $\mathbf{P}(X = r)$.
(b) Show that $\mathbf{E}(X) = np$.
(c) Suppose that $\mathbf{E}(X)$ is fixed, and let A be the event that $X \neq 0$. Find $\mathbf{E}(X|A)$, and show that as $n \to \infty$, $\mathbf{E}(X|A) \to \mathbf{E}(X)/(1 - \exp[-\mathbf{E}(X)])$.

Solution (a) We provide two solutions.
I Because characters are misprinted independently, the probability that r given characters are misprinted and the remaining $n - r$ are correct is $p^r(1 - p)^{n-r}$. Because there are $\binom{n}{r}$ distinct ways of fixing the positions of the r misprints, it follows that

$$\mathbf{P}(X = r) = \binom{n}{r} p^r (1 - p)^{n-r}. \tag{1}$$

Remark This is the *binomial* distribution, which we met in Example 4.2.3. We sometimes denote it by $\mathrm{B}(n, p)$.
II Consider the first character, and let M be the event that it is misprinted. Then,

$$\mathbf{P}(X = r) = \mathbf{P}(X = r|M)\mathbf{P}(M) + \mathbf{P}(X = r|M^c)\mathbf{P}(M^c).$$

We write $\mathbf{P}(X = r) = p(n, r)$ and observe that if M occurs, then $X = r$ if and only if there are $r - 1$ misprints in the remaining $n - 1$ characters. Hence,

$$p(n, r) = p(n - 1, r - 1)p + p(n - 1, r)(1 - p) \tag{2}$$

where $p(n, 0) = (1 - p)^n$ and $p(n, n) = p^n$, $n \geq 0$. Now the substitution $p(n, r) = p^r(1 - p)^{n-r}c(n, r)$ gives $c(n, r) = c(n - 1, r - 1) + c(n - 1, r)$, where $c(n, 0) = c(n, n) = 1$, $n \geq 0$.

We already know that this difference equation has the solution $c(n, r) = \binom{n}{r}$, as required (recall Pascal's triangle).

(b) We consider two solutions.
I Let $m(n)$ be the expected number of misprints in n characters. Then, by Theorem 4.4.6, $m(n) = \mathbf{E}(X|M)\mathbf{P}(M) + \mathbf{E}(X|M^c)\mathbf{P}(M^c) = (1 + m(n - 1))p + m(n - 1)(1 - p)$,

where we have used

$$\mathbf{E}(X|M) = \sum_r r\mathbf{P}(X = r|M) = \sum_r rp(n-1, r-1),$$

because misprints are independent,

$$= \sum_r p(n-1, r-1) + \sum_r (r-1)p(n-1, r-1) = 1 + m(n-1).$$

Hence,

$$m(n) = m(n-1) + p.$$

Obviously, $m(0) = 0$, so this difference equation has solution

$$m(n) = np. \tag{3}$$

II Using (a)

$$m(n) = \sum_{r=0}^{n} r\binom{n}{r}p^r(1-p)^{n-r} = \sum_{r=1}^{n} np\frac{(n-1)!}{(r-1)!(n-r)!}p^{r-1}(1-p)^{n-r}$$

$$= np\sum_{r=1}^{n}\binom{n-1}{r-1}p^{r-1}(1-p)^{n-r} = np \qquad \text{by the binomial theorem.}$$

(c) By definition,

$$\mathbf{E}(X|A) = \sum_{r=1}^{n}\frac{\mathbf{P}(X=r)}{\mathbf{P}(X>0)} = np/(1-(1-p)^n)$$

$$= np/\left(1-\left(1-\frac{np}{n}\right)^n\right) = \mathbf{E}(X)/\left(1-\left(1-\frac{\mathbf{E}(X)}{n}\right)^n\right)$$

$$\to \mathbf{E}(X)/(1-\exp(-\mathbf{E}(X)))$$

as $n \to \infty$.

(4) Exercise Show that $\text{var}(X) = np(1-p)$ by two different methods.

(5) Exercise Show that as $n \to \infty$, if $\mathbf{E}(X)$ is fixed $\mathbf{P}(X = 0) \to \exp(-\mathbf{E}(X))$.

(6) Exercise Let X have the binomial distribution $\mathbf{P}(X = k) = \binom{n}{k}p^k(1-p)^{n-k}$.

(a) For fixed n and p, for what value of k is $\mathbf{P}(X = k)$ greatest?

(b) For fixed k and p, for what value of n is $\mathbf{P}(X = k)$ greatest?

(7) Exercise If X has a binomial distribution with parameters n and p, find:

(a) The probability that X is even.

(b) $\mathbf{E}(\sin^2(\frac{1}{2}\pi X))$.

(c) $\mu_X^{(k)}$.

4.10 Example: Dog Bites: Poisson Distribution

(a) Let X be a binomial random variable with parameters n and p, such that $np = \lambda$. Show that for fixed k, as $n \to \infty$, with λ fixed,

$$\mathbf{P}(X = k) \to \frac{1}{k!}\lambda^k e^{-\lambda}.$$

(b) During 1979–1981, in Bristol, 1103 postmen sustained 215 dog bites. A total of 191 postmen were bitten, of whom 145 were bitten just once. Which should be the postman's motto: "Once bitten, twice shy" or "Once bitten, twice bitten"?

(8) Solution (a) Because X is binomial

$$\mathbf{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k} = \frac{n(n-1)\dots(n-k+1)}{n^k} \cdot \frac{\lambda^k}{k!} \cdot \left(1 - \frac{\lambda}{n}\right)^{n-k}.$$

Now for fixed k, as $n \to \infty$ with λ fixed,

$$\frac{n-k+j}{n} \to 1; \quad 1 \le j \le k,$$

$$\left(1 - \frac{\lambda}{n}\right)^k \to 1, \quad \text{and} \quad \left(1 - \frac{\lambda}{n}\right)^n \to e^{-\lambda}.$$

Hence, as $n \to \infty$,

$$\mathbf{P}(X = k) \to \frac{\lambda^k}{k!} e^{-\lambda}.$$

(4) Remark This is the *Poisson* distribution, which we met in Example 4.2.7.

(b) Suppose you were a postman, and let X be the number of your bites. If dogs bite any postman at random, then X is a binomial random variable with parameters 215 and $(1103)^{-1}$, because it may be thought of as a series of 215 trials in which a "success" is being bitten with probability $(1103)^{-1}$ at each trial, independently of the rest.

Hence,

$$\mathbf{P}(X = 0) = \left(1 - \frac{1}{1103}\right)^{215}$$

and

$$\mathbf{P}(X = 1) = \frac{215}{1103}\left(1 - \frac{1}{1103}\right)^{214}.$$

You may either compute these directly or recognise from (a) that the number of bites you get is approximately Poisson, with parameter $\lambda = \frac{215}{1103} \simeq 0.195$. So,

$$\mathbf{P}(X = 0) \simeq e^{-\lambda} \simeq 0.82$$
$$\mathbf{P}(X = 1) \simeq \lambda e^{-\lambda} \simeq 0.16$$
$$\mathbf{P}(X > 1) \simeq 1 - e^{-\lambda} - \lambda e^{-\lambda} \simeq 0.02.$$

However, if we pick a postman at random and let X' be the number of bites he sustained, we find that

$$\mathbf{P}(X' = 0) = \frac{912}{1103} \simeq 0.83$$
$$\mathbf{P}(X' = 1) = \frac{145}{1103} \simeq 0.13$$
$$\mathbf{P}(X' > 1) = \frac{46}{1103} \simeq 0.04.$$

It seems that "once bitten, twice bitten" should be the postman's motto.

(5) Remark Our conclusion may be given more substance by investigating the extent to which the observed distribution differs from the expected Poisson distribution. Such techniques are known to statisticians as "goodness-of-fit tests," and an appropriate procedure here would use the χ^2 test. This may be found in textbooks of elementary statistics; the motto is the same.

(1) Exercise If bites are Poisson with parameter λ, what is the probability that you get more than one bite, given that you get at least one?

(2) Exercise If bites are Poisson, what is your expected total number of bites given that you get at least one?

(3) Exercise Let X_n have a binomial distribution with parameters n and p, such that $np = \lambda$, and let A_n be the event that $X_n \geq 1$. If Y is a Poisson random variable with parameter λ, show that as $n \to \infty$, $\mathbf{P}(X_n = k | A_n) \to \mathbf{P}(Y = k | Y \geq 1)$.

(4) Exercise Let X have a Poisson distribution with parameter λ. Show that $\mathbf{E}(X) = \lambda$.
(a) For fixed λ, what value of k maximizes $\mathbf{P}(X = k)$?
(b) For fixed k, what value of λ maximizes $\mathbf{P}(X = k)$?

(5) Exercise If X has a Poisson distribution with parameter λ, find:
(a) $\mathbf{E}(e^X)$ (b) $\mathbf{E}(\cos(\pi X))$ (c) var (X) (d) $\mu_X^{(k)}$.

(6) Exercise If X has a Poisson distribution with parameter λ, show that:
(a) The probability that X is even is $e^{-\lambda}\cosh\lambda$.
(b) $\mathbf{E}(|X - \lambda|) = 2\lambda^{\lambda}e^{-\lambda}/(\lambda - 1)!$, when λ is an integer greater than zero.

4.11 Example: Guesswork

You are trying to guess the value of a proper integer valued random variable X, with probability mass function $f(x)$ (which you know). If you underestimate by y, it will cost you $\$by$; if you overestimate by y, it will cost you $\$ay$. Your guess is an integer; what guess minimizes your expected loss?

Solution If you guess t, then your expected loss is

$$L(t) = a\sum_{x \leq t}(t - x)f(x) + b\sum_{x > t}(x - t)f(x). \tag{1}$$

Substituting $t + 1$ for t in (1) gives an expression for $L(t + 1)$, and subtracting this from (1) gives

$$L(t) - L(t + 1) = a\sum_{x \leq t} f(x) + b\sum_{x > t} f(x) = -aF(t) + b(1 - F(t)). = D(t) \tag{2}$$

(say).

Now $\lim_{x \to -\infty} D(x) = b$, $\lim_{x \to \infty} D(x) = -a$, and both $-F(t)$ and $1 - F(t)$ are nonincreasing. Therefore, there is a smallest t such that

$$D(t) = L(t) - L(t + 1) \leq 0,$$

and this is the guess that minimizes your expected loss. Hence, denoting this guess by $\hat{t}$,

$$\hat{t} = \min\left\{t\colon F(t) \geq \frac{b}{a + b}\right\}, \quad \text{by (2).}$$

(3) Exercise Suppose that if you underestimate X you incur a fixed loss $£b$, whereas if you overestimate X by y it will cost you $£ay$. Find an expression that determines the guess that minimizes your expected loss.

Find this best guess when

(a) $\mathbf{P}(X = x) = pq^{x-1}; \quad x \geq 1, p = 1 - q > 0.$

(b) $\mathbf{P}(X = x) = 1/(x(x + 1)); \quad x \geq 1.$

(c) $\mathbf{P}(X = x) = 1/(2n + 1); \quad -n \leq x \leq n.$

(4) Exercise What is your best guess if

(a) $L(t) = \mathbf{E}(|X - t|)$?

(b) $L(t) = \mathbf{E}((X - t)^2)$?

(5) Exercise Icarus Airways sells $m + n$ tickets for its n-seat aeroplane. Passengers fail to show up with probability p independently. Empty seats cost $\$c$, and a passenger with a ticket who cannot fly is paid $\$b$ for being bumped. What choice of m minimizes the airline's expected losses on booking errors?

What level of compensation b would be sufficient to ensure that it was not worthwhile for the airline to overbook at all (for fixed p)? For fixed b, what value of p would entail no overbooking by the airline?

4.12 Example: Gamblers Ruined Again

Alberich and Brunnhilde have a and b gold coins, respectively. They play a series of independent games in which the loser gives a gold piece to the winner; they stop when one of them has no coins remaining. If Alberich wins each game with probability p, find the expected number of games played before they stop. (Assume $p \neq q = 1 - p$.)

(3) Solution Let X_k be the number of games they will play when Alberich's fortune is k, and let $m_k = \mathbf{E}(X_k)$. Clearly, $m_0 = m_{a+b} = 0$ because in each case one player has no coins. If A is the event that Alberich wins the first game, then for $0 < k < a + b$,

$$\mathbf{E}(X_k|A) = 1 + \mathbf{E}(X_{k+1}) = 1 + m_{k+1}$$

because his fortune is then $k + 1$, and succeeding games are independent of A. Likewise, it follows that

$$\mathbf{E}(X_k|A^c) = 1 + \mathbf{E}(X_{k-1}).$$

Hence, by Theorem 4.4.6,

$$m_k = \mathbf{E}(X_k|A)\mathbf{P}(A) + \mathbf{E}(X_k|A^c)\mathbf{P}(A^c) = 1 + pm_{k+1} + qm_{k-1}.$$

Setting

$$m_k = \frac{k}{q - p} + u_k,$$

gives

$$u_k = pu_{k+1} + qu_{k-1}, \quad \text{for } 0 < k < a + b.$$

In particular $u_0 = 0$, and $u_{a+b} = -(a+b)(q-p)^{-1}$. Proceeding as in Example 2.11, using Theorem 2.3.1, shows that

$$u_k = +\frac{(a+b)}{p-q}\,\frac{1-\left(\frac{q}{p}\right)^k}{1-\left(\frac{q}{p}\right)^{a+b}}. \tag{1}$$

(2) Exercise What is the expected number of games played when $p = \frac{1}{2}$?

(3) Exercise Let B be the event that Brunnhilde wins the entire contest. Find a difference equation satisfied by $\mathbf{E}(X_k|B)$. Solve this in the case when $p = \frac{1}{2}$.

(4) Exercise When the first game is over they redivide the $a+b$ coins as follows. All the coins are tossed, one player gets those showing a head, the other gets all those showing a tail. Now they play a series of games as before. What is the expected number to be played until one or other player again has all the coins? What if $p = \frac{1}{2}$?

(5) Exercise Alberich is blackmailing Fafner, so each time he loses his last gold coin, he immediately demands (and gets) one replacement coin, with which to continue gambling. What now is the expected number of games played? What if $p = \frac{1}{2}$?

4.13 Example: Postmen

A and B are postmen. They start work on day 1. The probability that A sustains a dog bite on day n, given that he has not been bitten on any of the preceding days is $p_A(n)$. The corresponding probability for B is $p_B(n)$. Let X_A and X_B, respectively, be the number of days until each sustains his first bite.

(a) Find $\mathbf{P}(X_A = n)$ and $\mathbf{P}(X_B = n)$.
(b) A is wary, so $p_A(n)$ decreases as n increases. If

$$p_A(n) = \frac{1}{n+1}; \quad n \geq 1,$$

find $\mathbf{P}(X_A = n)$ and show that $\mathbf{E}(X_A) = \infty$, while $\mathbf{P}(X_A < \infty) = 1$.
(c) B is complacent, so $p_B(n)$ increases as n increases. If, for some $\lambda < 0$,

$$p_B(n) = 1 - e^{-\lambda n}$$

find $\mathbf{P}(X_B = n)$ and $\mathbf{E}(X_B)$.

(2) Solution (a) Let H_k be the event that A is bitten on the kth day. Then the event that he is bitten for the first time on the nth day is $\bigcap_1^{n-1} H_k^c \cap H_n$. Hence,

$$\begin{aligned}\mathbf{P}(X_A = n) &= \mathbf{P}\left(\bigcap_1^{n-1} H_k^c \cap H_n\right) = \mathbf{P}\left(H_n | \bigcap_1^{n-1} H_k^c\right)\mathbf{P}\left(\bigcap_1^{n-1} H_k^c\right) \\ &= p_A(n)\mathbf{P}\left(H_{n-1}^c | \bigcap_1^{n-2} H_k^c\right)\mathbf{P}\left(\bigcap_1^{n-2} H_k^c\right) = p_A(n)\prod_1^{n-1}(1 - p_A(k)),\end{aligned}$$

on iterating.

Likewise, $\mathbf{P}(X_B = n) = p_B(n) \prod_1^{n-1} (1 - p_B(k))$.

(b) Employing a similar argument

$$\mathbf{P}(X_A > n) = \mathbf{P}\left(\bigcap_{k=1}^{n} H_k^c\right) = \prod_{k=1}^{n}(1 - p_A(k))$$
$$= \prod_{k=1}^{n}\left(1 - \frac{1}{k+1}\right) = \frac{1}{n+1} \to 0 \quad \text{as } n \to \infty.$$

Hence, $\mathbf{P}(X_A < \infty) = 1$. Also,

$$\mathbf{P}(X_A = n) = \mathbf{P}(X_A > n - 1) - \mathbf{P}(X_A > n) = \frac{1}{n(n+1)},$$

and finally $\mathbf{E}(X_A) = \sum_1^\infty 1/(n+1)$, which diverges. The expected time until A is first bitten is infinite.

(c) By the same argument,

$$\mathbf{P}(X_B \geq n) = \prod_{j=1}^{n}(1 - p_B(j)) = \prod_{j=1}^{n} e^{-\lambda j} = e^{-(\lambda/2)n(n+1)}.$$

Hence, $\mathbf{P}(X_B = n) = (1 - e^{-\lambda n})e^{-(\lambda/2)n(n-1)}$ and

$$\mathbf{E}(X_B) = \sum_1^\infty e^{-(\lambda/2)n(n+1)} < \infty.$$

B expects to be bitten in a finite time.

(1) **Exercise** In both cases (b) and (c), find the probability that the postman is first bitten on the jth day, given that he is bitten on or before day M.

(2) **Exercise** If A is less wary, so that $p_A(n) = 2/(n+2)$, show that $\mathbf{E}(X_A)$ is now finite, but var (X_A) diverges.

(3) **Exercise** In each case (b) and (c), given that the postman has not been bitten during the first m days, find the expected further time until he is bitten.

(4) **Exercise** If A is extremely wary and $p_A(n) = 1/(n+1)^2$ show that with probability $\frac{1}{2}$, he is never bitten. What is the median of the distribution of X_A in this case? Find the expectation of X_A, given that X_A is finite.

4.14 Example: Acme Gadgets

This company has developed a new product. The demand for it is unknown, but it is assumed to be a random variable X, which is distributed uniformly on $\{0, 1, \ldots, \mathrm{N}\}$. The gadgets have to be made in advance; each one sold makes a profit of $\$b$, and each one made and left unsold represents a net loss of $\$c$.

How many should be made, to maximize the expected profit?

Solution Suppose that m items are made. Then the total profit (negative profits are interpreted as losses) is

$$Y_m = \begin{cases} bm; & X \geq m \\ bX - c(m - X); & m > X \end{cases}.$$

The expected profit is

$$\begin{aligned} \mathbf{E}(Y_m) &= bm\mathbf{P}(X \geq m) + b\sum_{x=1}^{m-1} \frac{x}{N+1} - c\sum_{x=0}^{m-1} \frac{m-x}{N+1}, \\ &= \frac{m}{N+1}\left(b\left(N + \frac{1}{2}\right) - \frac{1}{2}c - \frac{1}{2}m(b+c)\right). \end{aligned}$$

Now,

$$2(N+1)(\mathbf{E}(Y_{m+1}) - \mathbf{E}(Y_m)) = (2N+1)b - c - (2m+1)(b+c),$$

so that the expected profit is largest when $m = \hat{m}$, where

$$\hat{m} = \max\left\{0, \left\lceil \frac{Nb - c}{b + c} \right\rceil\right\}.$$

(1) Exercise Suppose that an unsatisfied customer represents a loss of $\$d$. What now is the choice of m which maximizes expected profit?

(2) Exercise Suppose that the unknown demand X is assumed to have a geometric distribution with parameter p. Find the choice of m that maximizes the expected profit.

(3) Exercise Suppose the unknown demand X is a Poisson random variable with parameter λ. Show that the expected profit if they make m items is

$$\lambda(b+c) - \frac{(b+c)\lambda^{m+1}}{m!} \Big/ \sum_0^m \frac{\lambda^k}{k!} - mc,$$

and that this is maximized by the value of m that minimizes

$$\frac{(b+c)\lambda^{m+1}}{m!} \Big/ \sum_0^m \frac{\lambda^k}{k!} + mc.$$

4.15 Example: Roulette and the Martingale

Suppose you are playing roulette; the wheel has a zero. The chance of winning on red is $p < \frac{1}{2}$ and you bet at evens; if you win, you gain an amount equal to your stake.

Your first bet is \$1 on red. If it wins, you quit; if it loses, your second bet is \$2 on red. If it wins you quit, and so on. Your nth bet is $\$2^{n-1}$ so long as you lose; as soon as you win you quit.

(a) Show that you are certain to win \$1 every time you use this system.
(b) Find the expected size of your winning bet.

Now suppose the house limit is $\$2^L$, so this must be your last bet if you have not already won.

(c) What is your expected gain when you stop?
(d) Would you prefer large or small house limits?

Remark This gambling system is the martingale. Avoid it unless you seek ruin!

Solution Let T be the number of spins of the wheel until the outcome is first red.

(a) Your bet on that spin is $\$2^{T-1}$, and because you win, you gain $\$2^{T-1}$. However, your losses on the previous $T-1$ spins are $\$\sum_1^{T-1} 2^{k-1} = \$2^{T-1} - 1$. Because

$$\sum_{k=1}^{\infty} \mathbf{P}(T = k) = \sum_{k=1}^{\infty} p(1-p)^{k-1} = 1,$$

this means you are certain to win \$1.

(b) Because your winning bet is $\$2^{T-1}$, it has expected value

$$\mathbf{E}(2^{T-1}) = \sum_{k=1}^{\infty} 2^{k-1} p(1-p)^{k-1} = \infty, \quad \text{since } 2(1-p) > 1.$$

(c) You win \$1 if $1 \leq T \leq L+1$; otherwise, you lose $\$\sum_{k=0}^{L} 2^k$. Hence, your expected gains are $\$\gamma$, where

$$\gamma = \mathbf{P}(T \leq L+1) - (2^{L+1} - 1)\mathbf{P}(T > L+1) = 1 - (2(1-p))^{L+1}.$$

(d) Because your expected losses increase exponentially fast with L, you must hope the casino is sufficiently generous to have low limits.

(1) Exercise What difference does it make to these results if the wheel is fair? (That is, $p = \frac{1}{2}$.)

(2) Exercise With house limit $\$2^L$, what is the expected size of your winning bet, given that you do indeed win? What happens as $L \to \infty$? (Remember to consider all three cases, $p > \frac{1}{2}$, $p = \frac{1}{2}$, $p < \frac{1}{2}$.)

4.16 Example: Searching

(a) Let X be a positive integer valued random variable such that $f(n) = \mathbf{P}(X = n)$ is nonincreasing as n increases. Suppose that $(g(x); x = 1, 2, \ldots)$ is a function, taking positive integer values, such that for any k, $g(x) = k$ for at most one positive integer $x = r_k$.

Show that $\mathbf{E}(g(X)) \geq \mathbf{E}(X)$.

(b) You have lost a key. There are n places in which you might have mislaid it with respective probabilities $(p_k; 1 \leq k \leq n)$. If you search the kth place once, you find the key with probability d_k, if it is indeed there. (You can search any place any number of times.) How do you arrange your searching to minimize the expected time until you find the key? (Searches are successful independently of each other.)

Solution (a) Consider the distribution of $g(X)$. Because $g(X) = k$ for at most one value r_k of X,

$$\mathbf{P}(g(X) \leq n) = \sum_k \mathbf{P}(g(X) = k) = \sum_k \mathbf{P}(X = r_k) = \sum_k f(r_k)$$

where the final sum contains $m \leq n$ nonzero terms. If these are arranged in decreasing order as $f(r_{k_1}) \geq f(r_{k_2}) \geq \cdots \geq f(r_{k_m})$, then

$$f(r_{k_1}) \leq f(1)$$
$$f(r_{k_2}) \leq f(2),$$

and so on. Hence, summing these inequalities yields

$$\mathbf{P}(g(X) \leq n) \leq \mathbf{P}(X \leq n),$$

and so

$$\mathbf{E}(g(X)) = \sum_0^\infty \mathbf{P}(g(X) > n) = \sum_0^\infty (1 - \mathbf{P}(g(X) \leq n))$$
$$\geq \sum_0^\infty (1 - \mathbf{P}(X \leq n)) = \mathbf{E}(X).$$

(b) The probability that you find the key on the sth search of the rth room is

$$m_{rs} = (1 - d_r)^{s-1} d_r p_r.$$

To see this, note that the key has to be there (with probability p_r) and you have to fail to find it $s - 1$ times before you succeed. Let p_k be the kth largest of the numbers $(m_{rs}; r \geq 1, s \geq 1)$. Then p_k is a probability mass function and $(p_k; k \geq 1)$ is nonincreasing. Take this ordering as an order of search; that is, if $m_{rs} \geq m_{uv}$, then the sth search of the rth place precedes the vth search of the uth place. This search is consistent [the mth search of a given place precedes the $(m + 1)$st for every m], and $\sum kp_k$ is the expected number of searches required to find the key.

By part (a), any other order yields greater expected duration of the search time, because the function $g(x)$ is a permutation, and thus one–one.

(1) **Exercise** Show that you can arrange your searches so that the expected time to find the key is finite.

(2) **Exercise** The key is upstairs with probability $\frac{2}{3}$ or downstairs, with probability $\frac{1}{3}$. Any search upstairs is successful with probability $\frac{1}{4}$ if the key is there; any search downstairs is successful with probability $\frac{3}{4}$ if the key is there. How do you arrange your searches to minimize the expected number of searches?

(3) **Exercise** Suppose the sth search of the rth room (conditional on $s - 1$ previous unsuccessful searches of this room) discovers the key with probability d_{rs}. How do you order your searches to minimize the expected number of searches?

4.17 Example: Duelling

Pascal and Brianchon fight a series of independent bouts. At each bout, either Pascal is awarded a hit with probability p, or Brianchon is awarded a hit with probability $q = 1 - p$. The first to be awarded two consecutive hits is declared the winner and the duel stops. Let X be the number of bouts fought. Find the distribution and expected value of X. For what value of p is $\mathbf{E}(X)$ greatest?

Solution Let B be the event that Brianchon wins. Then,

$$f_X(n) = \mathbf{P}(\{X = n\} \cap B) + \mathbf{P}(\{X = n\} \cap B^c).$$

For B to occur at the nth bout, he must win the nth and $(n-1)$th bouts (with probability q^2), and the preceding $n-2$ bouts must be awarded alternately to each contestant. The probability of this is $p^{(n/2)-1}q^{(n/2)-1}$ if n is even, or $p^{(n/2)-(1/2)}q^{(n/2)-(3/2)}$ if n is odd, because bouts are independent.

A similar argument applies if B^c occurs, yielding

$$f_X(n) \begin{cases} p^{(n/2)-1}q^{(n/2)-1}(q^2+p^2) & \text{if } n \text{ is even} \\ p^{(n/2)-(1/2)}q^{(n/2)-(1/2)}(q+p) & \text{if } n \text{ is odd.} \end{cases}$$

The expected value of X is then, by definition,

$$\mathbf{E}(X) = \sum_{j=1}^{\infty} p^{j-1}q^{j-1}(q^2+p^2)2j + \sum_{j=1}^{\infty} q^j p^j (q+p)(2j+1).$$

Summing this series is elementary and boring. To get a solution in closed form, it is more fun to argue as follows. Let A_k be the event that Pascal is awarded the kth bout. Then,

$$\mathbf{E}(X) = \mathbf{E}(X|A_1)p + \mathbf{E}\big(X|A_1^c\big)q, \tag{1}$$

by conditioning on the outcome of the first bout. Now if Pascal is awarded the first bout but not the second, the state of the duel in respect of the final outcome is exactly the same as if he had lost the first bout, except of course that one bout extra has been fought. Formally, this says

$$\mathbf{E}\big(X|A_1 \cap A_2^c\big) = 1 + \mathbf{E}\big(X|A_1^c\big).$$

Hence,

$$\mathbf{E}(X|A_1) = \mathbf{E}(X|A_1 \cap A_2)p + \mathbf{E}\big(X|A_1 \cap A_2^c\big)q = 2p + q\big(1 + \mathbf{E}\big(X|A_1^c\big)\big). \tag{2}$$

Likewise,

$$\mathbf{E}\big(X|A_1^c\big) = 2q + p(1 + \mathbf{E}(X|A_1)). \tag{3}$$

Solving (2) and (3), and substituting into (1), yields

$$\mathbf{E}(X) = \frac{2+qp}{1-qp}.$$

Because $qp = \frac{1}{4} - (p - \frac{1}{2})^2$, this is greatest when $p = \frac{1}{2}$ and then $\mathbf{E}(X) = 3$.

(4) Exercise What is $\mathbf{P}(B)$?
(5) Exercise What is $\mathbf{P}(B^c)$?
(6) Exercise Given that Pascal wins, find the distribution and expected value of the number of bouts.
(7) Exercise Find $\mathbf{P}(A_1|B)$ and $\mathbf{P}(A_2|B)$.
(8) Exercise Find the median number of bouts when $p = \frac{1}{2}$.
(9) Exercise Find $\mathbf{P}(B)$ and the expectation of the number of bouts fought if the winner is required to win *three* consecutive bouts.

(10) Exercise Brianchon suggests that they adopt a different rule for deciding the winner, *viz*: when first a player has been awarded a total number of bouts two greater than the number of bouts awarded to his opponent, then the match stops and the leading player wins. If $p > q$, do you think Brianchon was wise to suggest this? (Assume he wants to win.) What is the expected duration of this game when $p = q$?

4.18 Binomial Distribution: The Long Run

Let X have a binomial distribution with parameters n and p, where $p = 1 - q$. Show that for $\lambda > 0$ and $\epsilon > 0$,

$$\mathbf{P}(X - np > n\epsilon) \leq \mathbf{E}(\exp[\lambda(X - np - n\epsilon)]).$$

Deduce that as $n \to \infty$,

$$\mathbf{P}(|X - np| \leq n\epsilon) \to 1.$$

(You may assume without proof that for any x, $0 < e^x \leq x + e^{x^2}$.)

Solution For $k > np + n\epsilon$, when $\lambda > 0$, we have $\exp(\lambda(k - np - n\epsilon)) > 1$. Hence

$$\begin{aligned}\sum_{k>n(p+\epsilon)} \mathbf{P}(X = k) &< \sum_{k>n(p+\epsilon)} \exp(\lambda(k - np - n\epsilon))\mathbf{P}(X = k) \\ &< \sum_{k} \exp(\lambda(k - np - n\epsilon))\mathbf{P}(X = k), \quad \text{because } e^x > 0, \\ &= \mathbf{E}(\exp(\lambda(X - np - n\epsilon))).\end{aligned}$$

Now, the left side is just $\mathbf{P}(X > np + n\epsilon)$ and

$$\mathbf{E}(e^{\lambda X}) = \sum_0^n \binom{n}{k} (pe^{\lambda})^k q^{n-k} = (q + pe^{\lambda})^n,$$

so the right side is

$$\begin{aligned}(pe^{\lambda q} + qe^{-\lambda p})^n e^{-\lambda n\epsilon} &\leq (pe^{\lambda^2 q^2} + qe^{\lambda^2 p^2})^n e^{-\lambda n\epsilon}, \quad \text{because } e^x \leq x + e^{x^2} \\ &\leq \exp(n\lambda^2 - \lambda n\epsilon).\end{aligned}$$

Now, choosing $\lambda = \epsilon/2$ gives

$$\mathbf{P}(X - np > n\epsilon) \leq \exp(-n\epsilon^2/4).$$

Likewise, $\mathbf{P}(X - np < n\epsilon) \leq \exp(-n\epsilon^2/4)$, so

$$\mathbf{P}(|X - np| > n\epsilon) \leq 2\exp(-n\epsilon^2/4) \to 0 \quad \text{as } n \to \infty, \tag{1}$$

as required.

(2) Exercise You want to ask each of a large number n of people a question to which the answer "yes" is so embarrassing that many individuals would falsely answer "no". The answer "no" is not embarrassing. The following procedure is proposed to determine the embarrassed fraction of the population. As the question is asked, a coin is tossed out of sight of the questioner. If the true answer would have been "no" and the coin shows heads, then the answer "yes" is given. Otherwise, people

should respond truthfully. If the number responding "yes" is now Y_n and "yes" is the true answer for a proportion p of the whole population, show that for $\epsilon > 0$

$$\mathbf{P}\left(\left|\frac{Y_n}{n} - \frac{1}{2}(1+p)\right| > \epsilon\right) \leq 2\exp(-n\epsilon^2/4).$$

Explain the advantages of this procedure.

(3) Exercise Suppose a coin shows a head with probability p, and let X_n be the number of heads in n tosses, and $A_n(\epsilon)$ the event that $|X_n/n - p| > \epsilon$, where $2 > \epsilon > 0$. Show that as $n \to \infty$,

$$\mathbf{P}\left(\bigcup_n^{\infty} A_k(\epsilon)\right) \to 0.$$

(4) Exercise Suppose a gambler wins \$28 with probability $\frac{1}{32}$, or loses his stake of \$1 with probability $\frac{31}{32}$ at each trial. Let W_n be his fortune after n such independent trials, and $A_n(\epsilon)$ the event that $|W_n/n + 3/32| > \epsilon$. Show that as $n \to \infty$, $\mathbf{P}(\bigcup_{n=m}^{\infty} A_n(\epsilon)) \to 0$. Deduce that his fortune is equal to its initial value on only finitely many occasions, with probability one. (*Hint*: recall Problem 1.24.)

Note: In the following exercise, X is a binomially distributed random variable with parameters n and p.

(5) Exercise Show that for any fixed finite a and b, as $n \to \infty$, $\mathbf{P}(a < X \leq b) \to 0$.

(6) Exercise Show that for $a > 0$,

$$\mathbf{P}\left(\left|\frac{X}{n} - p\right| > a\right) \leq \frac{(p(1-p))^{1/2}}{a^2 n} \min\{(p(1-p))^{1/2}, an^{1/2}\}.$$

(7) Exercise (a) Show that if $p = (m-1)/n$ where m is an integer, then

$$\mathbf{E}\left(\left|\frac{X}{n} - p\right|\right) = 2\binom{n-1}{m-1} p^m (1-p)^{n-m+1}.$$

(b) Find var $(|X/n - p|)$.

(8) Exercise If $n = 2m$ and $p = \frac{1}{2}$, show that

$$\mathbf{P}(X - m = k) = \binom{2m}{m} \frac{1}{4^m} a(m,k)$$

where, as $m \to \infty$, $(a(m,k))^m \to e^{-k^2}$.
Also, show that

$$\frac{1}{2m^{\frac{1}{2}}} < \binom{2m}{m} \frac{1}{4^m} < \frac{1}{(2m+1)^{\frac{1}{2}}}.$$

[You may assume that $|\log(1+x) - x| < x^2$ for small enough x.]

4.19 Example: Uncertainty and Entropy

Let X and Y be simple random variables taking values in the same set $\{x_1, \ldots, x_n\}$, with respective probability mass functions $f_X(.)$ and $f_Y(.)$. Show that

(1) $$-\mathbf{E}(\log f_Y(X)) \geq -\mathbf{E}(\log f_X(X)),$$

and

$$-\mathbf{E}(\log f_X(X)) \le \log n, \tag{2}$$

with equality in (1) if and only if $f_Y(.) \equiv f_X(.)$, and equality in (2) if and only if $f_X(x_i) = n^{-1}$ for all x_i. (*Hint*: Show first that $\log x \le x - 1$.)

Solution By definition, for $x > 0$,

$$-\log x = \int_x^1 y^{-1}\, dy \ge \int_x^1 dy = 1 - x, \tag{3}$$

with equality if and only if $x = 1$. Hence,

$$\begin{aligned}
\mathbf{E}(\log f_X(X)) - \mathbf{E}(\log f_Y(X)) &= \sum_i f_X(x_i) \log f_X(x_i) - \sum_i f_X(x_i) \log f_Y(x_i) \\
&= -\sum_i f_X(x_i) \log[f_Y(x_i)/f_X(x_i)] \\
&\ge -\sum_i f_X(x_i)[1 - f_Y(x_i)/f_X(x_i)] \quad \text{by (3)} \\
&= 0,
\end{aligned}$$

with equality iff $f_X(x_i) = f_Y(x_i)$ for all x_i, which proves (1). In particular, setting $f_Y(x_i) = n^{-1}$ yields (2).

Remark It is conventional to denote $-\mathbf{E}(\log f_X(X))$ by $H(X)$ [or alternatively $h(X)$] and the logarithms are taken to base 2. The number $H(X)$ is known as the *uncertainty* or *entropy* of X, and is an essential tool in information theory and communication theory. The result (1) is sometimes called the Gibbs inequality.

(4) Exercise Let $f_X(x) = \binom{n}{x} p^x (1-p)^{n-x}$; $0 \le x \le n$. Show that

$$H(X) \le -n(p \log p + (1-p)\log(1-p)),$$

with equality if $n = 1$.

(5) Exercise Let $f_X(x) = pq^{x-1}/(1-q^M)$, for $1 \le x \le M$, where $p = 1 - q$. Show that

$$\lim_{M \to \infty} H(X) = -p^{-1}[p \log p + (1-p)\log(1-p)].$$

(6) Exercise Let $Y = g(X)$ be a function of the random variable X. Show that for any $c > 0$

$$H(Y) \le H(X) \le c\mathbf{E}(Y) + \log\left(\sum_i e^{-cg(x_i)}\right).$$

When does equality hold?

PROBLEMS

1 A box contains 12 sound grapefruit and four that are rotten. You pick three at random.
(a) Describe the sample space.
(b) Let X be the number of sound grapefruit you pick. Find $f_X(x)$ and $\mathbf{E}(X)$.

2 Show that the expected number of pairs in your poker hand is about 0.516.

3 You roll a die once. What is the variance of your score?

4 What is the variance of a uniform random variable?

5 For each of the following functions $f(x)$ (defined on the positive integers $x = 1, 2, \ldots$), find:

(a) The value of c for which $f(x)$ is a probability mass function.

(b) The expectation

(i) $f(x) = c.2^x/x!$ (iv) $f(x) = cx^{-2}$

(ii) $f(x) = cp^x; \quad 0 \le p \le 1$ (v) $f(x) = c[x(x+1)]^{-1}$

(iii) $f(x) = cp^x x^{-1}; \quad 0 \le p \le 1$

6 If X is a random variable, explain whether it is true that $X + X = 2X$ and $X - X = 0$. Are 0 and $2X$ random variables?

7 For what value of c is $f(x) = c(x(x+1)(x+2))^{-1}; 1 \le x \le M$, a probability mass function? Find its expectation $\mathbf{E}(X)$. Find the limit of c and $\mathbf{E}(X)$ as $M \to \infty$.

8 A fair coin is tossed repeatedly. Let A_n be the event that three heads have appeared in consecutive tosses for the first time on the nth toss. Let T be the number of tosses required until three consecutive heads appear for the first time. Find $\mathbf{P}(A_n)$ and $\mathbf{E}(T)$.

Let U be the number of tosses required until the sequence *HTH* appears for the first time. Can you find $\mathbf{E}(U)$?

9 You choose a random number X as follows. Toss a coin repeatedly and count the number of tosses until it shows a head, N say. Then pick an integer at random in $1, 2, \ldots, 10^N$. Show that

$$\mathbf{P}(X = k) = \frac{1}{19} \cdot \frac{1}{20^{d-1}},$$

where d is the number of digits in the decimal expansion of k. What is $\mathbf{E}(X)$?

10 Let X have a Poisson distribution $f(k)$, with parameter λ. Show that the largest term in this distribution is $f([\lambda])$.

11 Show that if $\mathbf{E}(X^2) < \infty$, $\min_a \mathbf{E}((X-a)^2) = \text{var}(X)$.

12 Let $f_1(x)$ and $f_2(x)$ be probability mass functions. Show that if $0 \le p \le 1$, then $f_3(x) = pf_1(x) + (1-p)f_2(x)$ is a probability mass function. Interpret this result.

13 Let X be a geometric random variable. Show that, for $n > 0$ and $k > 0$, $\mathbf{P}(X > n + k | X > n) = \mathbf{P}(X > k)$.

14 Let X be a random variable uniform on $1 \le x \le m$. What is $\mathbf{P}(X = k | a \le X \le b)$? In particular find $\mathbf{P}(X > n + k | X > n)$.

15 A random variable is symmetric if for some a and all k, $f(a-k) = f(a+k)$. Show that the mean and a median are equal for symmetric random variables. Find a nonsymmetric random variable for which the mean and median are equal.

16 If X is symmetric about zero and takes integer values, find $\mathbf{E}(\cos(\pi X))$ and $\mathbf{E}(\sin(\pi X))$.

17 Let X have distribution function F. Find the distribution of $Y = aX + b$ and of $Z = |X|$.

18 Let X have a geometric distribution such that $\mathbf{P}(X = k) = q^{k-1}p; \; k \ge 1$. Show that $\mathbf{E}(X^{-1}) = \log(p^{(1/p-1)})$.

19 (a) Let X have a Poisson distribution with parameter λ. Show that $\mathbf{E}(1/(X+1)) = \lambda^{-1}(1 - e^{-\lambda})$, and deduce that for all λ, $\mathbf{E}(1/(X+1)) \ge (\mathbf{E}(X+1))^{-1}$. When does equality hold?

(b) Find $\mathbf{E}(1/(X+1))$ when $\mathbf{P}(X = k) = (-k^{-1}p^k)/\log(1-p); \; k \ge 1$.

20 **Fingerprints** It is assumed that the number X of individuals in a population, whose fingerprints are of a given type, has a Poisson distribution with some parameter λ.

(a) Explain when and why this is a plausible assumption.

(b) Show that $\mathbf{P}(X = 1 | X \ge 1) = \lambda(e^\lambda - 1)^{-1}$.

(c) A careless miscreant leaves a clear fingerprint of type t. It is known that the probability that any randomly selected person has this type of fingerprint is 10^{-6}. The city has 10^7 inhabitants and a citizen is produced who has fingerprints of type t. Do you believe him to be the miscreant on this evidence alone? In what size of city would you be convinced?

21 Initially urn I contains n red balls and urn II contains n blue balls. A ball selected randomly from urn I is placed in urn II, and a ball selected randomly from urn II is placed in urn I. This whole operation is repeated indefinitely. Given that r of the n balls in urn I are red, find the mass function of R, the number of red balls in urn I after the next repetition.

Show that the mean of this is $r + 1 - 2r/n$, and hence find the expected number of red balls in urn I in the long run.

22 A monkey has a bag with four apples, three bananas, and two pears. He eats fruit at random until he takes a fruit of a kind he has eaten already. He throws that away and the bag with the rest. What is the mass function of the number of fruit eaten, and what is its expectation?

23 **Matching** Consider the matching problem of Example 3.17. Let $\mu^{(k)}$ be the kth factorial moment of the number X of matching letters, $\mu^{(k)} = \mathbf{E}(X(X-1)\dots(X-k+1))$. Show that

$$\mu^{(k)} = \begin{cases} 1; & k \le n \\ 0; & k > n. \end{cases}$$

24 Suppose an urn contains m balls which bear the numbers from 1 to m inclusive. Two balls are removed with replacement. Let X be the difference between the two numbers they bear.

(a) Find $\mathbf{P}(X \le n)$.

(b) Show that if $n/m = x$ is fixed as $m \to \infty$, then $\mathbf{P}(|X| \le n) \to 1 - (1-x)^2$; $0 \le x \le 1$.

(c) Show that $\mathbf{E}|X|/m \to \frac{1}{3}$.

25 Suppose the probability of an insect laying n eggs is given by the Poisson distribution with mean $\mu > 0$, that, is by the probability distribution over all the nonnegative integers defined by $p_n = e^{-\mu}\mu^n/n!\,(n = 0, 1, 2, \dots)$, and suppose further that the probability of an egg developing is p. Assuming mutual independence of the eggs, show that the probability distribution q_m for the probability that there are m survivors is of the Poisson type and find the mean.

26 Preparatory to a camping trip, you can buy six cans of food, all the same size, two each of meat, vegetables, and fruit. Assuming that cans with the same contents have indistinguishable labels, in how many distinguishable ways can the cans be arranged in a row?

On the trip, there is heavy rain and all the labels are washed off. Show that if you open three of the cans at random the chance that you will open one of each type is $\frac{2}{5}$. If you do not succeed, you continue opening cans until you have one of each type; what is the expected number of open cans?

27 A belt conveys tomatoes to be packed. Each tomato is defective with probability p, independently of the others. Each is inspected with probability r; inspections are also mutually independent. If a tomato is defective and inspected, it is rejected.

(a) Find the probability that the nth tomato is the kth defective tomato.

(b) Find the probability that the nth tomato is the kth rejected tomato.

(c) Given that the $(n+1)$th tomato is the first to be rejected, let X be the number of its predecessors that were defective. Find $\mathbf{P}(X = k)$, the probability that X takes the value k, and $\mathbf{E}(X)$.

28 Mr. Smith must site his widget warehouse in either Acester or Beeley. Initially, he assesses the probability as p that the demand for widgets is greater in Acester, and as $1 - p$ that it is greater in Beeley. The ideal decision is to site the warehouse in the town with the larger demand. The cost of the wrong decision, because of increased transport costs, may be assumed to be £1000 if Acester is the correct choice and £2000 if Beeley is the correct choice. Find the expectations of these costs

for each of the two possible decisions, and the values of p for which Acester should be chosen on the basis of minimum expected cost.

Mr. Smith could commission a market survey to assess the demand. If Acester has the higher demand, the survey will indicate this with probability $\frac{3}{4}$ and will indicate Beeley with probability $\frac{1}{4}$. If Beeley has the higher demand the survey will indicate this with probability $\frac{2}{3}$ and will indicate Acester with probability $\frac{1}{3}$. Show the probability that the demand is higher in Acester is $9p/(4+5p)$ if the survey indicates Acester. Find also the expected cost for each of the two possible decisions if the survey indicates Acester.

If the survey indicates Acester and $p < 8/17$, where should Mr. Smith site the warehouse?

29 A coin is tossed repeatedly and, on each occasion, the probability of obtaining a head is p and the probability of obtaining a tail is $1-p$ $(0 < p < 1)$.

(a) What is the probability of not obtaining a tail in the first n tosses?

(b) What is the probability p_n of obtaining the first tail at the nth toss?

(c) What is the expected number of tosses required to obtain the first tail?

30 The probability of a day being fine is p if the previous day was fine and is p' if the previous day was wet. Show that, in a consecutive sequence of days, the probability u_n that the nth is fine satisfies $u_n = (p-p')u_{n-1} + p'$, $n \geq 2$.

Show that as $n \to \infty$, $u_n \to p'(1-p+p')^{-1}$.

By considering the alternative possibilities for tomorrow's weather, or otherwise, show that if today is fine the expected number of future days up to and including the next wet day is $1/(1-p)$.

Show that (today being fine) the expected number of future days up to and including the next two consecutive wet days is $(2-p)/((1-p)(1-p'))$.

31 Cars are parked in a line in a parking lot in order of arrival and left there. There are two types of cars, small ones requiring only one unit of parking length (say 15 ft) and large ones requiring two units of parking length (say 30 ft). The probability that a large car turns up to park is p and the probability that a small car turns up is $q = 1-p$. It is required to find the expected maximum number of cars that can park in a parking length of n units, where n is an integer. Denoting this number by $M(n)$ show that:

(a) $M(0) = 0$

(b) $M(1) = 1-p$

(c) $M(n) - qM(n-1) - pM(n-2) = 1, \quad (n \geq 2)$

Show that the equations are satisfied by a solution of the form $M(n) = A\alpha^n + B\beta^n + Cn$, where α, β are the roots of the equation $x^2 - qx - p = 0$, and A, B, C are constants to be found. What happens to $M(n)$ as $n \to \infty$?

32 The probability that the postman delivers at least one letter to my house on any day (including Sundays) is p. Today is Sunday, the postman has passed my house and no letter has been delivered.

(a) What is the probability that at least one letter will be delivered during the next week (including next Sunday)?

(b) Given that at least one letter is delivered during the next week, let X be the number of days until the first is delivered. What is $f_X(x)$?

(c) What is the expected value of X?

(d) Suppose that all the conditions in the first paragraph hold, except that it is known that a letter will arrive on Thursday. What is the expected number of days until a letter arrives?

33 A gambler plays two games, in each of which the probability of her winning is 0.4. If she loses a game she loses her stake, but if she wins she gets double her stake. Suppose that she stakes a in the first game and b in the second, with $a+b=1$. Show that her expected loss after both games is 0.2.

Suppose she plays again, but now the stake in the first game buys knowledge of the second, so that the chance of winning in the second is ap (≤ 1). Show that the value of a which gives the greatest expected gain is $0.5 + 0.2/p$.

34 Let $f_1(X)$ and $f_2(X)$ be functions of the random variable X. Show that (when both sides exist) $[\mathbf{E}(f_1 f_2)]^2 \leq \mathbf{E}(f_1^2)\mathbf{E}(f_2^2)$. Deduce that $\mathbf{P}(X = 0) \leq 1 - [\mathbf{E}(X)]^2/\mathbf{E}(X^2)$.
(Recall that $at^2 + 2bt + c$ has distinct real roots if and only if $b^2 > ac$.)

35 Any oyster contains a pearl with probability p independently of its fellows. You have a tiara that requires k pearls and are opening a sequence of oysters until you find exactly k pearls. Let X be the number of oysters you have opened that contain no pearl.
(a) Find $\mathbf{P}(X = r)$ and show that $\sum_r \mathbf{P}(X = r) = 1$.
(b) Find the mean and variance of X.
(c) If $p = 1 - \lambda/k$, find the limit of the distribution of X as $k \to \infty$.

36 A factory produces 100 zoggles a day. Each is defective independently with probability p. If a defective zoggle is sold, it costs the factory £100 in fines and replacement charges. Therefore, each day 10 are selected at random and tested. If they all pass, all 100 zoggles are sold. If more than one is defective, then all 100 zoggles are scrapped. If one is defective, it is scrapped and a further sample of size 10 is taken. If any are defective, the day's output is scrapped; otherwise, 99 zoggles are sold.
(a) Show that the probability r of not scrapping the day's output is $(1 - p)^{10}(1 + 10p(1 - p)^9)$.
(b) If testing one zoggle costs £10, find the expected cost of a day's testing.
(c) Find the expected returns on a day's output in terms of the profit b of a sold zoggle and cost c of a scrapped zoggle.

37 An urn contains two blue balls and $n - 2$ red balls; they are removed without replacement.
(a) Show that the probability of removing exactly one blue ball in $r - 1$ removals is

$$\frac{2(r-1)(n-r+1)}{n(n-1)}.$$

(b) Show that the probability that the urn first contains no blue balls after the rth removal is

$$\frac{2(r-1)}{n(n-1)}.$$

(c) Find the expected number of removals required to remove both blue balls.

38 Suppose that n dice are rolled once; let X be the number of sixes shown. These X dice are rolled again, let Y be the number of sixes shown after this second set of rolls.
(a) Find the distribution and mean of Y.
(b) Given that the second set of rolls yielded r sixes, find the distribution and mean of X.

39 Pascal and Brianchon now play a series of games that may be drawn (i.e., tied) with probability r. Otherwise, Pascal wins with probability p or Brianchon wins with probability q, where $p + q + r = 1$.
(a) Find the expected duration of the match if they stop when one or other wins two consecutive games. Also, find the probability that Pascal wins.
(b) Find the expected duration of the match if they stop when one or other wins two successive games of the games that are won. (That is, draws are counted but ignored.) Find the probability that Pascal wins.
If you were Brianchon and $p > q$, which rules would you rather play by?

40 Let the random variable X have a geometric distribution, $\mathbf{P}(X = k) = q^{k-1}p; k \geq 1$. Show that for $t > 0$, $\mathbf{P}(X \geq a + 1) \leq pe^{-ta}(1 - qe^t)^{-1}$. Deduce that $\mathbf{P}(X \geq a + 1) \leq (a + 1)p[q(a + 1)a^{-1}]^a$, and compare this with the exact value of $\mathbf{P}(X \geq a + 1)$.

41 An archer shoots arrows at a circular target of radius 1 where the central portion of the target inside radius $\frac{1}{4}$ is called the bull. The archer is as likely to miss the target as she is to hit it. When the

archer does hit the target, she is as likely to hit any one point on the target as any other. What is the probability that the archer will hit the bull? What is the probability that the archer will hit k bulls in n attempts? Prove that the mean number of bulls that the archer hits in n attempts is $n/32$.

Show that if the archer shoots 96 arrows in a day, the probability of her hitting no more than one bull is approximately $4e^{-3}$. Show that the average number of bulls the archer hits in a day is 3, and that the variance is approximately $(63\sqrt{3}/64)^2$.

42 Prove Chebyshov's inequality that, for a random variable X with mean μ and variance σ^2,

$$\mathbf{P}(|X-\mu| \le h\sigma) \ge 1 - \frac{1}{h^2}, \quad \text{for any } h > 0.$$

When an unbiased coin is tossed n times, let the number of tails obtained be m. Show that

$$\mathbf{P}\left(0.4 \le \frac{m}{n} \le 0.6\right) \ge 0.75$$

when $n \ge 100$. Given that $n = 100$, show that

$$\mathbf{P}\left(0.49 \le \frac{m}{n} \le 0.51\right) \simeq 3(5\sqrt{(2\pi)})^{-1}.$$

(You may assume Stirling's formula that $n! \simeq \sqrt{(2\pi)}n^{n+1/2}e^{-n}$ when n is large.)

43 An ambidextrous student has a left and a right pocket, each initially containing n humbugs. Each time she feels hungry she puts a hand into one of her pockets and if it is not empty, takes a humbug from it and eats it. On each occasion, she is equally likely to choose either the left or the right pocket. When she first puts her hand into an empty pocket, the other pocket contains H humbugs.

Show that if p_h is the probability that $H = h$, then

$$p_h = \binom{2n-h}{n} \frac{1}{2^{2n-h}},$$

and find the expected value of H, by considering $\sum_{h=0}^{n}(n-h)p_h$, or otherwise.

44 You insure your car. You make a claim in any year with probability q independently of events in other years. The premium in year j is a_j (where $a_j < a_k$ for $k < j$), so long as no claim is made. If you make a claim in year k, then the premium in year $k + j$ is a_j as long as no further claim is made, and so on. Find the expected total payment of premiums until the first claim.

45 A Scotch die has faces bearing tartan patterns: three are McDiarmid, two are Meldrum, and one is Murray. Show that the expected number of times you must roll the die before all three patterns have appeared is 7.3.

46 **Tail Sums** Let $X \ge 0$ be integer valued. Use the indicator $I(X > k)$ to prove that

$$\mathbf{E}X = \sum_{k \ge 0} \mathbf{P}(X > k),$$

and

$$\mathbf{E}X^r = \sum_{k \ge 0} rk^{r-1}\mathbf{P}(X > k).$$

47 **Coupon Collecting: Example (4.3.15) Revisited** Let X_n be the number of coupons collected until you first obtain a coupon that is a duplicate of one you already possess. Find $\mathbf{P}(X_n = k)$ and deduce that

(a) $\displaystyle\sum_{k=2}^{n+1} \frac{n!}{(n-k+1)!}\frac{k-1}{n^k} = 1.$

(b) $\mathbf{E}X_n = \displaystyle\sum_{k=0}^{n} \frac{n!}{(n-k)!}n^{-k}.$

48 Let $(x_i; 1 \leq i \leq n)$ be a collection of positive numbers. Show that

$$\left(\frac{1}{n}\sum_{i=1}^{n}\frac{1}{x_i}\right)^{-1} \leq \left(\prod_{i=1}^{n} x_i\right)^{1/n}.$$

49 If $(y_i; 1 \leq i \leq n)$ is any ordering of $(x_i; 1 \leq i \leq n)$, show that $\sum_{i=1}^{n} \frac{x_i}{y_i} \geq n$.

50 Let X have finite variance, and set $v(x) = \mathbf{E}(X - x)^2$. Show that $\mathbf{E}v(X) = 2\text{var}X$.

51 Let X have mean μ, variance σ^2, and median m. Use (4.6.4) to show that $|\mu - m| < \sigma$.

5
Random Vectors: Independence and Dependence

Wherever there is wealth there will be dependence and expectation.
Samuel Johnson [The Rambler, 189]

5.1 Joint Distributions

Commonly, each outcome of an experiment generates two (or more) real numbers of interest. We can treat these as individual random variables $(X_i; 1 \leq i \leq n)$, but it is often important to consider their joint behaviour. For example, if the experiment is your visit to your doctor, you may find out your height H and weight W. These are separate random variables, but are often informative when considered jointly. Thus, the outcome $H = 150\,\text{cm}$ and $W = 150\,\text{kg}$ might disturb your physician, whereas the outcome $H = 190\,\text{cm}$ and $W = 80\,\text{kg}$ probably would not. Likewise, the random vector comprising height, weight, age, sex, blood pressure, and heart rate is of more use considered jointly than separately.

As another example, complicated systems (e.g., space shuttles) have several on-board computers that work together to run the system. If one fails or makes an error, the others can override it; thus, the system fails only when a majority of the computers fail. If X_i is the time until the ith processor fails, then the time until the system fails depends jointly on the collection of random variables, $X_1, \ldots, X_n$. It is natural to refer to such a collection as a random vector, and write $X = (X_1, X_2, \ldots, X_n)$. Formally, as before, we have $X = X(\omega); \omega \in \Omega$ and $A_x = \{\omega : X(\omega) = x\} \in \mathcal{F}$, but we do not often refer to the underlying sample space Ω. Because X maps Ω into a countable subset S of R^n, we think of S as the sample space. (You may well have already been doing this instinctively in Chapter 4.)

For simplicity, we summarize the properties of random vectors in two dimensions; the appropriate generalizations in more dimensions are straightforward.

Definition *Let X and Y be two discrete random variables taking values $(x_i; i \geq 1)$ and $(y_j; i \geq 1)$, respectively. Their* joint probability mass function *$f(x, y)$ is defined by*

$$f(x, y) = \mathbf{P}(X = x, Y = y)$$

as x and y range over all possible values x_i and y_j of X and Y. ▲

The mass function $f(x, y)$ is zero, except at a countable set of points in $\mathbb{R}^2$. In fact,

$$f(x, y) \geq 0 \quad \text{for all } x \text{ and } y, \tag{1}$$

and further if $\sum_{i,j} f(x_i, y_j) = 1$, then the joint distribution $f(x, y)$ is not defective. Most important is the Key Rule:

$$\mathbf{P}((X, Y) \in C) = \sum_{(x,y)\in C} f(x, y). \tag{2}$$

(3) Example Suppose that a coin is tossed twice; let X be the total number of heads shown and Y the total number of tails. Then, (X, Y) takes values in

$$S = \{0, 1, 2\} \times \{0, 1, 2\} = \{(i, j) : i \in \{0, 1, 2\}, j \in \{0, 1, 2\}\}.$$

Clearly, $f(x, y)$ is zero, except at the points $(0, 2)$, $(1, 1)$, and $(2, 0)$. Furthermore,

$$f(0, 2) + f(1, 1) + f(2, 0) = (1 - p)^2 + 2p(1 - p) + p^2 = 1,$$

where we have denoted the probability of a head by p, as usual. ●

Any real function $g(X, Y)$ of two such jointly distributed random variables is itself a random variable. If we set $Z = g(X, Y)$, then Z has a probability mass function given by using the Key Rule (2) above:

$$f_Z(z) = \mathbf{P}(g(X, Y) = z) = \sum f(x, y), \tag{4}$$

where the summation is over all x and y, such that $g(x, y) = z$.

In particular, if $g(x, y) = x$, we have

$$f_X(x) = \sum_y f(x, y), \tag{5}$$

and if $g(x, y) = y$, we have

$$f_Y(y) = \sum_x f(x, y). \tag{6}$$

Thus, we have shown the important result that, if we know the joint mass function of several random variables, we can find all their separate mass functions. When obtained in this way, $f_X(x)$ and $f_Y(y)$ are sometimes called *marginal* mass functions.

Here are some examples illustrating joint mass functions.

(7) Example A row of n numbered machines are producing components that are identical, except for the serial number. On any day, the kth component produced by the jth machine bears the serial number (j, k). On the day in question, the rth machine produces c_r $(1 \leq r \leq n)$ components, and at the end of the day one component C is picked at random from all those produced. Let its serial number be (X, Y). Find $f(x, y)$, $f_X(x)$ and $f_Y(y)$.

Solution Because C is picked at random from all $\sum_{r=1}^{n} c_r$ components, we have

$$f(x, y) = \left(\sum_{1}^{n} c_r \right)^{-1}, \quad 1 \le x \le n; 1 \le y \le c_x,$$
$$= a \quad \text{(say)}.$$

Then, by (5),

$$f_X(x) = \sum_y f(x, y) = ac_x.$$

Now define the function $H(i, j) = \begin{cases} 1 & \text{if } c_i \ge c_j, \\ 0 & \text{otherwise.} \end{cases}$

Then, by (6),

$$f_Y(y) = \sum_x f(x, y) = a \sum_{x=1}^{n} H(x, y).$$

●

(8) Example: Cutting for the Deal It is customary, before engaging in a card game, to cut for the deal; each player removes a portion of the deck in turn, and then each reveals the bottom card of his segment. The highest card wins. For these to be random variables, we need to assign numerical values to the court cards, so we set J = 11, Q = 12, K = 13, A = 14, when aces are high.

(a) Art and Bart cut for deal, aces high. Let X be Art's card, and Y be Bart's card. Find the joint mass function of X and Y. Does it make any difference how many cards Art removes from the deck?
(b) Let V be the loser's card, and W the dealer's (winning) card. Find the joint mass function of V and W, and the separate mass functions of V and W.
(c) Find the mass function of the dealer's winning margin (namely, $W - V$).
(d) What is the mass function of the dealer's card when three players cut for deal?

Note that in the event of a tie, the deck is shuffled and the players cut again to choose the dealer.

Solution (a) Each random variable takes values in $\{2, 3 \ldots, 14\}$. Cutting the deck twice amounts to selecting two cards at random, and because ties are not allowed, $X \ne Y$. By symmetry, any two unequal values are equally likely to occur, so

$$f_{X,Y}(x, y) = \mathbf{P}(X = x, Y = y) = \begin{Bmatrix} \frac{1}{12} \cdot \frac{1}{13} & x \ne y \\ 0 & x = y \end{Bmatrix} \quad 2 \le x, y \le 14.$$

It makes no difference how many cards Art removes with his cut.

(b) Of course, $W > V$, so

$$f_{V,W}(v, w) = \mathbf{P}(V = v, W = w) = \mathbf{P}(X = v, Y = w) + \mathbf{P}(X = w, Y = v)$$
$$= \begin{cases} \frac{2}{12} \cdot \frac{1}{13} = \frac{1}{78}; & 2 \le v < w \le 14 \\ 0 \cdot \qquad\quad ; & \text{otherwise.} \end{cases}$$

This is otherwise obvious, because the experiment amounts to choosing an unordered pair of unequal cards at random, with equal probability of choosing any pair. Hence, for $v < w$, $f(v, w) = \binom{13}{2}^{-1}$, as above. Now, by (5),

$$f_V(v) = \sum_{w=v+1}^{14} \frac{1}{78} = \frac{14 - v}{78}; \quad 2 \leq v \leq 13.$$

Then by (6),

$$f_W(w) = \sum_{2}^{w-1} \frac{1}{78} = \frac{w - 2}{78}; \quad 3 \leq w \leq 14.$$

(c) By (4),

$$f_Z(z) = \mathbf{P}(W - V = z) = \sum \frac{1}{78},$$

where the summation is over all v and w such that $w - v = z$. Because $z \leq v < w \leq 14$, there are exactly $13 - z$ terms in this sum, so

$$\mathbf{P}(W - V = z) = \frac{13 - z}{78}; \quad 1 \leq z \leq 12.$$

(d) Arguing as we did for (b), where now $u < v < w$, we have

$$\mathbf{P}(U = u, V = v, W = w) = \binom{13}{3}^{-1}.$$

Hence,

$$\begin{aligned} f_W(w) &= \sum_{2 \leq u < v < w} \binom{13}{3}^{-1}; \quad 4 \leq w \leq 14, \\ &= \sum_{3 \leq v < w} (v - 2)\binom{13}{3}^{-1} = \frac{1}{2}(w - 3)(w - 2)\binom{13}{3}^{-1}. \end{aligned}$$ ●

(9) Example For what value of c is the function

$$f(x, y) = c\binom{x + y - 1}{x}\lambda^x \mu^y, \quad x \geq 0; y \geq 1; 1 > 1 - \lambda > \mu > 0$$

a joint mass function? For this value of c, find the mass functions of X and Y.

Solution By (2),

$$\begin{aligned} c^{-1} &= \sum_{y=1}^{\infty} \sum_{x=0}^{\infty} \binom{x + y - 1}{x}\lambda^x \mu^y = \sum_{y=1}^{\infty} \frac{\mu^y}{(1 - \lambda)^y} \quad \text{by (3.6.12),} \\ &= \frac{\mu}{1 - \lambda - \mu}. \end{aligned}$$

Then, by (5),

$$f_X(x) = c\sum_{y=1}^{\infty} \mu\lambda^x \binom{x+1+y-2}{y-1}\mu^{y-1} = \frac{(1-\lambda-\mu)\lambda^x}{(1-\mu)^{x+1}}, \quad x \geq 0.$$

Likewise,

$$f_Y(y) = c\sum_{x=0}^{\infty} \binom{x+y-1}{x}\lambda^x\mu^y = \frac{(1-\lambda-\mu)\mu^{y-1}}{(1-\lambda)^y}, \quad y \geq 1.$$

Thus, $X+1$ and Y are both geometric, with parameters $\frac{\lambda}{1-\mu}$ and $\frac{\mu}{1-\lambda}$, respectively, X taking values in the nonnegative integers and Y in the positive integers. ●

(10) Example Leif and Rolf are bored with fair games. They want to play a game in which the probability of winning (for Leif) is λ, where λ is an arbitrary number in $[0,1]$. Also, they want the game to be of finite duration with probability 1. Unfortunately, the only gaming aid they have is a fair coin. Can you supply them with a game?

Solution Let λ have binary expansion

$$\lambda = 0.b_1b_2b_3\ldots = \sum_{n=1}^{\infty} b_n 2^{-n}.$$

Now toss the coin repeatedly and let I_n be the indicator of the event that the nth toss is a head. Let T be the first toss such that $I_n \neq b_n$, $T = \min\{n\colon I_n \neq b_n\}$. If $I_T < b_T$, then Leif wins; otherwise, Rolf wins. Now, $\mathbf{P}(T=n) = (\frac{1}{2})^n$ so that

$$\mathbf{P}(T<\infty) = \sum_{1}^{\infty}\left(\frac{1}{2}\right)^n = 1$$

and indeed $\mathbf{E}(T) = 2$. Also, Leif can only win at the nth toss if $b_n = 1$ so

$$\mathbf{P}(\text{Leif wins}) = \sum_n b_n\mathbf{P}(T=n) = \sum_n b_n 2^{-n} = \lambda,$$

as required. ●

5.2 Independence

Given the joint mass function of X and Y, equations (5.1.5) and (5.1.6) yield the marginal mass functions of X and Y. However, to be given the marginal distributions does not in general uniquely determine a joint distribution.

(1) Example Let X and Y have joint mass function given by

$$f(0,0) = \frac{1}{6},\ f(0,1) = \frac{1}{3},\ f(1,0) = \frac{1}{12},\ f(1,1) = \frac{5}{12},$$

and let U and V have joint mass function given by

$$f(0,0) = \frac{1}{4},\ f(0,1) = \frac{1}{4},\ f(1,0) = 0,\ f(1,1) = \frac{1}{2}.$$

Then, summing to get the marginal mass functions shows that:

$$f_X(0) = f_U(0) = \frac{1}{2},\ f_X(1) = f_U(1) = \frac{1}{2};$$

and

$$f_Y(0) = f_V(0) = \frac{1}{4},\ \ f_Y(1) = f_V(1) = \frac{3}{4}.$$

These marginal mass functions are the same, but the joint mass functions are different. ●

There is one exceptionally important special case when marginal mass functions do determine the joint mass function uniquely.

(2) Definition *Random variables X and Y are independent if, for all x and y,*

$$f(x, y) = f_X(x) f_Y(y).$$

This is equivalent to $\mathbf{P}(A \cap B) = \mathbf{P}(A)\mathbf{P}(B)$, where $A = \{\omega\colon X(\omega) = x\}$ and $B = \{\omega\colon X(\omega) = y\}$, which is the definition of independence for the events A and B. More generally, a collection $(X_i; 1 \le i \le n)$ with mass function f is independent if for all $x = (x_1, \ldots, x_n)$

(3)
$$f(x) = \prod_{i=1}^{n} f_{X_i}(x_i).$$

▲

Note that if X or Y (or both) are improper random variables [so that $f(x, y)$ is defective], then to say they are independent is interpreted as meaning

(4)
$$\mathbf{P}(X = x, Y = y) = \mathbf{P}(X = x)\mathbf{P}(Y = y)$$

for all finite x and y. This may seem odd, but such random variables occur quite naturally in simple random walks and other topics.

Example 5.1.7 Revisited Recall that n machines produce components. Suppose that all the machines produce the same number c of components, and as before we pick one at random and let its serial number be (X, Y), where X is the machine number and Y is the component index. Then,

$$f(x, y) = (nc)^{-1}; \quad 1 \le x \le n;\ 1 \le y \le c$$

and

$$f_X(x) = \frac{1}{n}; \quad 1 \le x \le n \qquad \text{and} \qquad f_Y(y) = \frac{1}{c}; \quad 1 \le y \le c.$$

Obviously, $f(x, y) = f_X(x) f_Y(y)$ and so X and Y are independent in this case. ●

(5) Example 1 Revisited Observe that the mass functions of X and Y, and of U and V, do not satisfy Definition 2. Let W and Z be independent random variables such that $f_W(0) = \frac{1}{2}$, $f_W(1) = \frac{1}{2}$ and $f_Z(0) = \frac{1}{4}$, $f_Z(1) = \frac{3}{4}$.

Then, by Definition 2, their joint mass function is

$$f(0,0) = \frac{1}{8},\ f(0,1) = \frac{3}{8},\ f(1,0) = \frac{1}{8},\ f(1,1) = \frac{3}{8}. \quad \bullet$$

(6) Example 5.1.9 Revisited Observe that X and Y are not independent because $\binom{x+y-1}{x}\lambda^x\mu^y$ cannot be expressed in the form $f_X(x)f_Y(y)$. If $X + 1$ and Y were independent geometric random variables with parameters $\frac{\lambda}{1-\mu}$ and $\frac{\mu}{1-\lambda}$, then the joint mass function would be

$$f(x,y) = \left(1 - \frac{\lambda}{1-\mu}\right)\left(1 - \frac{\mu}{1-\lambda}\right)\left(\frac{\lambda}{1-\mu}\right)^x\left(\frac{\mu}{1-\lambda}\right)^{y-1}, \quad x \geq 0; y \geq 1.$$

$\bullet$

The apparently simple Definition 2 implies a great deal more about independent random variables, as the following result shows.

(7) Theorem *Let X and Y be independent random variables. Then*:
(a) *For arbitrary countable sets A and B,*

(8) $$\mathbf{P}(X \in A, Y \in B) = \mathbf{P}(X \in A)\mathbf{P}(Y \in B),$$

and
(b) *For any real functions $g(\cdot)$ and $h(\cdot)$, $g(X)$ and $h(Y)$ are independent.*

Proof (a) The left-hand side of (8) is

$$\begin{aligned}\sum_{x\in A}\sum_{y\in B}\mathbf{P}(X = x, Y = y) &= \sum_{x\in A}\sum_{y\in B}\mathbf{P}(X = x)\mathbf{P}(Y = y) \quad \text{by independence} \\ &= \sum_{x\in A}\mathbf{P}(X = x)\sum_{y\in B}\mathbf{P}(Y = y) = \mathbf{P}(X \in A)\mathbf{P}(Y \in B),\end{aligned}$$

as required. For (b), let $A = \{x\colon g(X) = \xi\}$ and $B = \{y\colon h(Y) = \eta\}$. Then, by part (a), for any ξ and η,

$$\begin{aligned}\mathbf{P}(g(X) = \xi, h(Y) = \eta) &= \mathbf{P}(X \in A, Y \in B) = \mathbf{P}(X \in A)\mathbf{P}(Y \in B) \\ &= \mathbf{P}(g(X) = \xi)\mathbf{P}(h(Y) = \eta),\end{aligned}$$

as required. ■

(9) Example Independent random variables X and Y take the values -1 or $+1$ only, and $\mathbf{P}(X = 1) = a$, $\mathbf{P}(Y = 1) = \alpha$. A third random variable Z is defined by $Z = \cos((X + Y)\frac{\pi}{2})$. If $0 < a, \alpha < 1$, show that there are unique values of a and α such that X and Z are independent, and Y and Z are independent. In this case, are X, Y, and Z independent?

Solution First, for Z,

$$\mathbf{P}(Z = 1) = \mathbf{P}(X + Y = 0) = a(1 - \alpha) + \alpha(1 - a)$$

and, likewise, $\mathbf{P}(Z = -1) = a\alpha + (1 - a)(1 - \alpha)$. Now,

$$\mathbf{P}(Z = 1, X = 1) = \mathbf{P}(X = 1, Y = -1) = a(1 - \alpha).$$

Hence, if

$$a(1 - \alpha) = a(a(1 - \alpha) + \alpha(1 - a)) \tag{10}$$

we have $\mathbf{P}(Z = 1, X = 1) = \mathbf{P}(Z = 1)\mathbf{P}(X = 1)$. Simplifying (10) yields $\alpha = \frac{1}{2}$. Now, plodding through three similar constraints shows that X and Z are independent iff $\alpha = a = \frac{1}{2}$. By symmetry, the same condition holds iff Y and Z are independent.

However, X, Y, and Z are not independent because

$$\mathbf{P}(X = 1, Y = 1, Z = -1) = 0 \neq \mathbf{P}(X = 1)\mathbf{P}(Y = 1)\mathbf{P}(Z = -1). \qquad \bullet$$

Independent random variables often have interesting and useful properties.

(11) Example Let X and Y be independent geometric random variables having respective mass functions $f_X(x) = (1 - \lambda)\lambda^x$ and $f_Y(y) = (1 - \mu)\mu^y$ for $x \geq 0$ and $y \geq 0$. What is the mass function of $Z = \min\{X, Y\}$?

Solution By independence,

$$\mathbf{P}(Z > n) = \mathbf{P}(X > n \cap Y > n) = \mathbf{P}(X > n)\mathbf{P}(Y > n) = \lambda^{n+1}\mu^{n+1} = (\lambda\mu)^{n+1}.$$

Hence,

$$\mathbf{P}(Z = n) = \mathbf{P}(Z > n - 1) - \mathbf{P}(Z > n) = (1 - \lambda\mu)(\lambda\mu)^n$$

and Z is also geometric with parameter $\lambda\mu$. $\bullet$

5.3 Expectation

Let the random variable $Z = g(X, Y)$ be a function of X and Y. Using (5.1.4) and the definition of expectation (4.3.1), we have

$$\mathbf{E}(Z) = \sum_z z f_Z(z) = \sum_z z\mathbf{P}(g(X, Y) = z).$$

This expression for $\mathbf{E}(Z)$ is not always simple or convenient for use in calculation. The following generalization of Theorem 4.3.4 is therefore very useful.

(1) Theorem *Let X and Y have joint mass function $f(x, y)$. Whenever the sum on the right-hand side is absolutely convergent, we have*

$$\mathbf{E}(g(X, Y)) = \sum_{x,y} g(x, y) f(x, y).$$

Proof The proof is essentially the same as that of Theorem 4.3.4. $\blacksquare$